

AKADEMIKA

DCL-QAM

**8 QAM/DQAM MODULATION
& DEMODULATION KIT
EXPERIMENTAL MANUAL**

.... A MESSAGE FROM

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We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

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SAFETY RULES

Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.

After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.

This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault, please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solution's sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the warranty period if the service is needed, purchaser should get in touch with the service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solution's judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solution bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solution does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solution is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or sales outlet.

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TECHNICAL SPECIFICATIONS

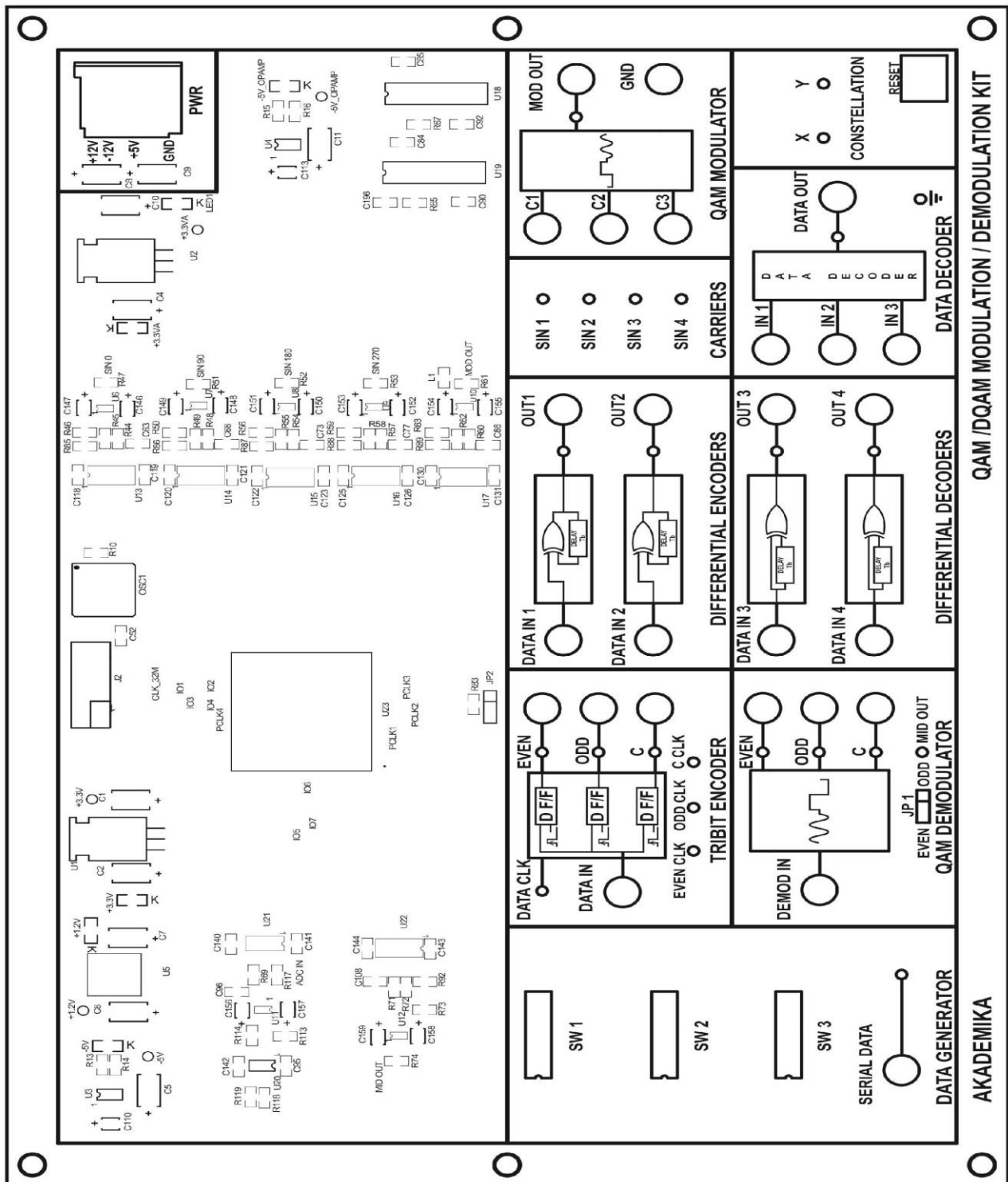
DCL-QAM: QAM/DQAM MODULATION AND DEMODULATION KIT

DATA SIMULATOR	: Onboard 24-bit variable NRZ-L pattern.
CRYSTAL OSCILLATOR	: 32.768 MHz
DATA CLOCK	: 256 KHz.
DATA ENCODING	: NRZ (L), Tribit data (EVEN, ODD & C). Differentially encoded EVEN & ODD
ONBOARD CARRIER SINE WAVES	: 1MHz (0°), 1MHz (90°), 1MHz (180°), 1MHz (270°)
CARRIER MODULATION	: QAM, DQAM.
CARRIER DEMODULATION	: QAM, DQAM.
DATA DECODING	: NRZ (L), Tribit data (EVEN, ODD & C). Differentially decoded EVEN& ODD
CONSTELLATION DIAGRAM	: Provision for observing constellation diagram and the concept can be well understood.
INTERMEDIATE SIGNAL	: Provision for observing intermediate signals during demodulation.
INTER CONNECTION	: 2 mm banana socket.
POWER SUPPLY	: +12V, -12V, +5V, GND.
TEST POINTS	: 30.

ACCESSORIES

DCL-QAM			Quantity
1	Short Links PCS2 Patch cord Male To Male		15
2	ADCL-05 EXPT Manual		1
3	5 PIN DIN CONNECTOR Power Supply		1
4	Power cord		1

FUNCTIONAL BLOCK DIAGRAM



INTRODUCTION

AKADEMIKA with the set of this trainer, namely, “DCL-QAM” initiates the user to the various Data Conditioning and Carrier Modulation Techniques, normally adopted in practice.

“DCL-QAM/DQAM” MODULATION AND DEMODULATION KIT

FEATURES:

- 1) The System illustrates the use of the following data formats:
 - a) Non Return to Zero - Level (NRZ-L),
 - b) Tribit encoded data
 - c) Differentially encoded EVEN & ODD.
- 2) The trainers come with an onboard data simulator, which generates the NRZ-L pattern depending on the positions of switches of 8-bit DIP switch and the reference clock (256 KHz), which enable the trainers to work in a standalone mode.
- 3) Two carrier modulation options are available on the kit.
 - a) Quadrature Amplitude Modulation. (QAM)
 - b) Differentially Quadrature Amplitude Modulation (DQAM).
- 4) Onboard carrier generation circuit generates sine waves synchronized to the transmitted data. The Four carrier sine waves generated onboard are of frequencies.
 - a) 1MHz (0°),
 - b) 1MHz (90°),
 - c) 1MHz (180°),
 - d) 1MHz (270°)
- 5) Following options are available for detecting the information from the carrier:
 - 1) “MID OUT” signals which are the output of the multipliers before filtering.
 - 2) QAM /DQAM Detection using “FIR FILTER”.
- 6) Interconnection facilities:
Sockets and connecting chords are provided for connections on board.
- 8) Test Points:
All relevant test points are brought-out for observations. Observations are carried out on an oscilloscope.

PRINCIPLES OF ADVANCED DIGITAL MODULATION AND DEMODULATION TECHNIQUES:

The Digital data in the systems can be encoded in several formats. All these Waveforms can be broadly classified into the following four groups:

1. Non-return to Zero formats,
2. Return to Zero formats,
3. Phase encoded formats,
4. Multilevel binary formats.

It is common that some puzzling questions may arise to a beginner as to why there should be so many PCM Waveforms? And are there really so many unique applications necessitating such a variety of waveforms to represent the “ones” and “zeros”. The reason for the large selection relates to the difference in performances that characterizes each waveform. In choosing a coding scheme for a particular application, the common parameters worth examination are as follows:

1) DC Component:

The DC component associated with the data formats has to be minimized, if the system is to be AC coupled. The proper selection of the data format ensures optimum DC component that is associated with the signal power spectrum.

2) Self Clocking:

The presence of sufficient number of transition in the transmitted data enables the receiver to derive the clock from the data. Thus it is preferable to select a data format, which enables sufficient number of transitions in the transmitted data stream. For example, the biphase signals allow sufficient number of transitions in the transmitted data, compared to the corresponding Non-return to zero signals.

3) Band Width:

The proper selection of the coding enables to optimize the bandwidth requirement.

4) Noise Immunity:

The various PCM waveform types can be further characterized by the probability of bit error versus signal to noise ratio. For Example, the NRZ waveforms have better immunity to noise than the corresponding unipolar return to zero signal, which enables an error free transmission.

FUNDAMENTALS OF CODING AND CARRIER MODULATION TECHNIQUES:

Digital Communication Systems represent the information by the binary digits, 'Ones' and 'Zeros'. The basic principles of digital communication are Sampling, Multiplexing, Encoding, Error Control Coding, Data Conditioning, and Carrier Modulation etc. In coding we represent sample data pattern in different formats. In carrier modulation, carrier is modulated as per modulating signal.

CARRIER MODULATION SCHEMES:

Digital Communication is the technique by which the whole information is represented in terms of binary digits, i.e., 'ones' and 'zeros'. These digits are represented by discrete voltage levels and the clock frequency of a digital communication scheme is generally low. For long distance transmission, the data is made to modulate a continuous wave (sine wave) carrier. These techniques are called Carrier Modulation Techniques. The various types of Carrier Modulation

Techniques normally adopted in practice fall under three broad categories.

- a) Amplitude Shift Keying (ASK),
- b) Frequency Shift Keying (FSK),
- c) Phase Shift Keying (PSK).

a) AMPLITUDE SHIFT KEYING (ASK):

For all types of Carrier Modulation, the Carrier frequency should be at least 2 times base band of the modulating signal.

Assuming that the modulated carrier is a sine wave represented by the following equation.

$$C(t) = A(t) \cdot \cos(wt)$$

Where C (t) = Carrier sine wave.
 A (t) = Time varying amplitude
 wt = Time varying angle.

Now in amplitude shift keying, the carrier is being transmitted only when the modulating data is 'one' and when the data is 'zero' the carrier is rejected from transmission. Thus the resulting modulated output of this type of modulation can be represented as follows:

$M(t) = r(t) \cdot C(t)$
 Where M (t) = Modulated Carrier
 r (t) = Time varying modulating data which is either 'one' or 'zero'

Now,

$$M(t) = A(t) \cdot \cos(wt) \text{ when modulating data is one.}$$

$$0 \text{ when modulating data is zero.}$$

An Envelope detector is used to recover the data from the modulated carrier.

b) FREQUENCY SHIFT KEYING (FSK):

In this type of modulation, the modulated output shifts between two frequencies for all 'one' to 'zero' transitions.

Let the carrier frequencies be represented by w_1 and w_2 , and then we have:

$$\begin{aligned} M(t) &= A(t) \cdot \cos(w_1 t) && \text{if data is 'One'} \\ &= A(t) \cdot \cos(w_2 t) && \text{if data is 'Zero'}. \end{aligned}$$

Where $A(t)$ = Time varying amplitude of the sine wave.

$M(t)$ = Modulated carrier.

FSK Demodulator employs PLL logic for the recovery of data.

c) PHASE SHIFT KEYING (PSK)

In the PSK modulation or phase shift keying, for all 'one' to 'zero' transitions of the modulating data, the modulated output switches between the in phase and out of phase components of the modulating frequency. If the modulated carrier is represented by:

$$\begin{aligned} M(t) &= A(t) \cdot \cos(wt + \text{Phase}) \\ \text{Where } A(t) &= \text{Time varying amplitude, and} \\ wt &= \text{Time varying angle.} \\ M(t) &= \text{Modulated carrier.} \end{aligned}$$

Then the phase equals 0° whenever the data equals 'one' and the phase equals 180° whenever the data equals 'zero'. This type of phase shift keying is called Binary phase Shift Keying (BPSK).

There are also other forms of phase shift keying like Differential Phase Shift Keying (DPSK), Differentially Encoded Phase Shift Keying (DEPSK) Quadrature Phase Shift Keying (QPSK), and Differential Quadrature Phase Shift Keying (DQPSK) etc.

In DPSK/DEPSK the data stream to be transmitted is encoded in such a way to get the differentially encoded data as input to DPSK/DEPSK modulator. Then the phase equals 0° whenever the data equals 'one' and the phase equals 180° whenever the data equals 'zero'. This type of phase shift keying is called Differential phase Shift Keying (DPSK)/ differentially encoded phase Shift Keying (DEPSK).

In QPSK, phase has 4 different values like 0° , 90° , 180° and 270° . They are used in the carrier modulation of differentially coded dibit pair. One of the four carriers is transmitted at a time with respect to the symbol generated by EVEN & ODD. In DQPSK the inputs to the modulator (EVEN & ODD) are differentially encoded before applying to the modulator & then transmitted similar to the QPSK. At the receiver recovered data decoded before applying to the data decoder.

The BPSK/DPSK/DEPSK detector works on the principle of square law.

BPSK/DPSK/DEPSK modulated carrier can be represented mathematically as:

$$\begin{aligned} M(t) &= A(t) \cdot \cos(wt + \text{Phase}) \\ \text{We have phase} &= 0 \text{ degree} && \text{- when data is 1} \\ &= 180 \text{ degrees} && \text{- when data is 0} \\ \text{Thus,} & & & \\ M(t) &= A(t) \cdot \cos(wt) && \text{- when data is 1} \\ &= -A(t) \cdot \cos(wt) && \text{- when data is 0} \end{aligned}$$

On squaring the modulated carrier, the negative sign gets eliminated and two multiplies the frequency.

$$\begin{aligned} M_2(t) &= A_2(t) \cdot \cos^2(\omega t) && \text{- for both data is 1 and 0} \\ M_2(t) &= A_2(t) \cdot (1 + \cos(2\omega t))/2 \end{aligned}$$

Now the in phase reference carrier can be recovered by dividing the frequency of the squared modulated carrier by two.

Once the carrier is recovered, the data can be detected by comparing the phase of the received modulated carrier with the phase of the reference carrier.

Quadrature Amplitude modulation (QAM) is a digital modulation where the information is contained into the phase as well as the amplitude of the transmitted carrier. As an example of a QAM system, let us consider that we propose to transmit a symbol for every 3-bit. There are then $2^3 = 8$ different possible symbols and we shall have to be able to generate 8 distinguishable signals. One possible representation of 8 signals is shown in fig. In this configuration each signal point is equally distant from its nearest neighbors, the distance being $d = 2a$.

Let us assume that all 8 signals are equally likely. Because of the symmetry displayed we can determine the average energy associated with a signal, from the four signals in the first quadrant. The average normalized energy of a signal is

$$\begin{aligned} E_s &= \frac{1}{4}[(a_2 + a_2) + (9a_2 + a_2) + (a_2 + 9a_2) + (9a_2 + 9a_2)] \\ &= 10a_2 \end{aligned} \quad \text{eq. 1}$$

So that

$$a = \sqrt{0.1 E_s}$$

And

$$d = 2\sqrt{0.1 E_s}$$

In the present case since each symbol represents 3 bits, the normalized symbol energy is $E_s = 3 E_b$ where E_b is the normalized bit energy. Hence

$$a = \sqrt{0.1 E_s} = \sqrt{0.4 E_b} \text{ and } d = 2\sqrt{0.4 E_b}$$

This distance is significantly less than the distance between adjacent QPSK signals which is $d = 2\sqrt{E_b}$; however the distance is greater or 16 QAM than for 8

MPSK which is

$$\begin{aligned} d &= \sqrt{8 E_b} \sin^2 \pi/8 \\ &= 2\sqrt{0.15 E_b} \end{aligned}$$

Thus, 8 QAM will be shown to have an error rate than 8 MPSK, but a higher error rate than QPSK.

A typical signal in eq.1 is

$$v_{QAM} = k_1 a u_1(t) + k_2 a u_2(t)$$

In which k_1 and k_2 are each equal to ± 1 or ± 3 . Since we have that

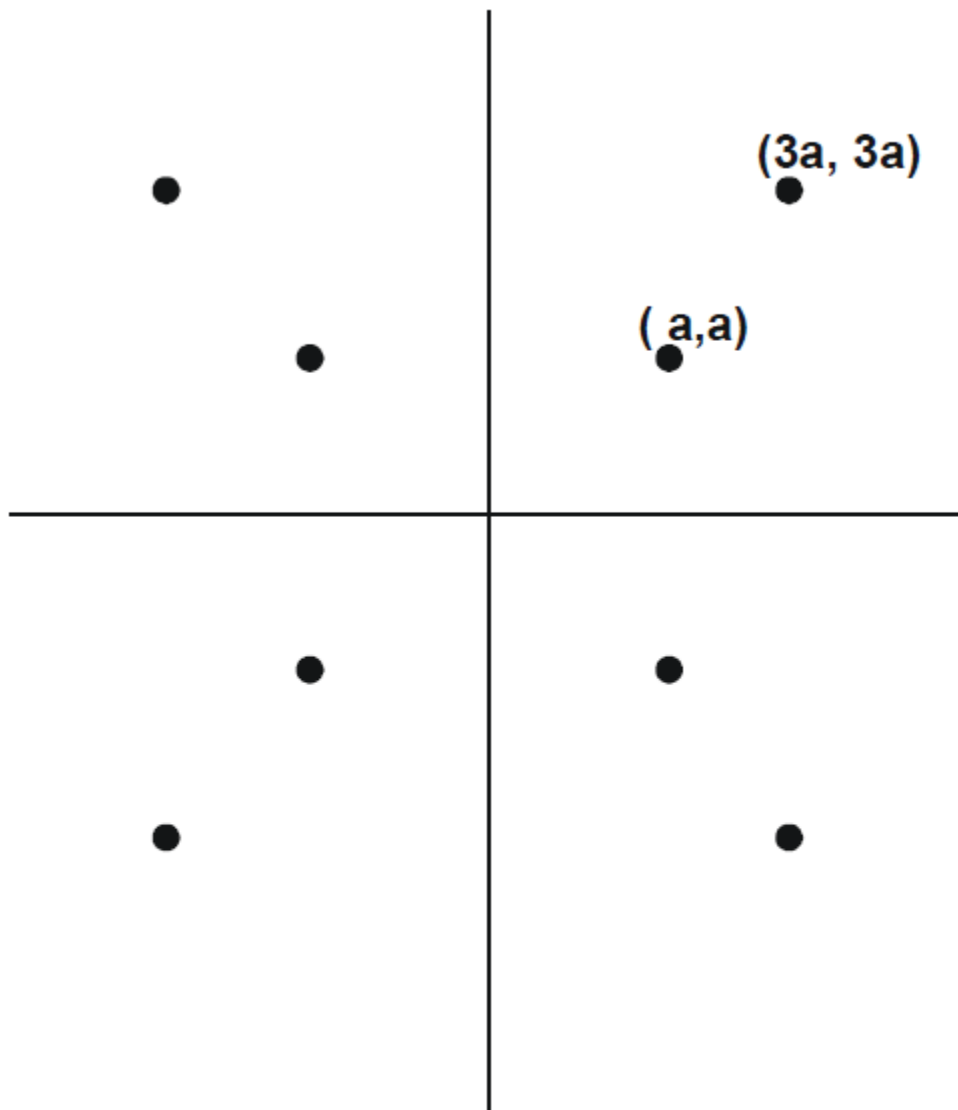
$$u_1(t) = \sqrt{(2/T_s)} \cos \omega_0 t, \text{ and that } u_2(t) = \sqrt{(2/T_s)} \sin \omega_0 t \text{ and } a = \sqrt{0.1 E_s}$$

We can write eq.2 as

$$v_{QAM} = k_1 \sqrt{0.2 E_s/T_s} \cos \omega_0 t + k_2 \sqrt{0.2 E_s/T_s} \sin \omega_0 t \quad \text{eq.2}$$

and since $E_s/T_s = P_s$ we have

$$v_{QAM} = k_1 (\sqrt{0.2 P_s}) \cos \omega_0 t + k_2 (\sqrt{0.2 P_s}) \sin \omega_0 t$$



8-QAM CONSTELLATION

EXPERIMENT NO.1

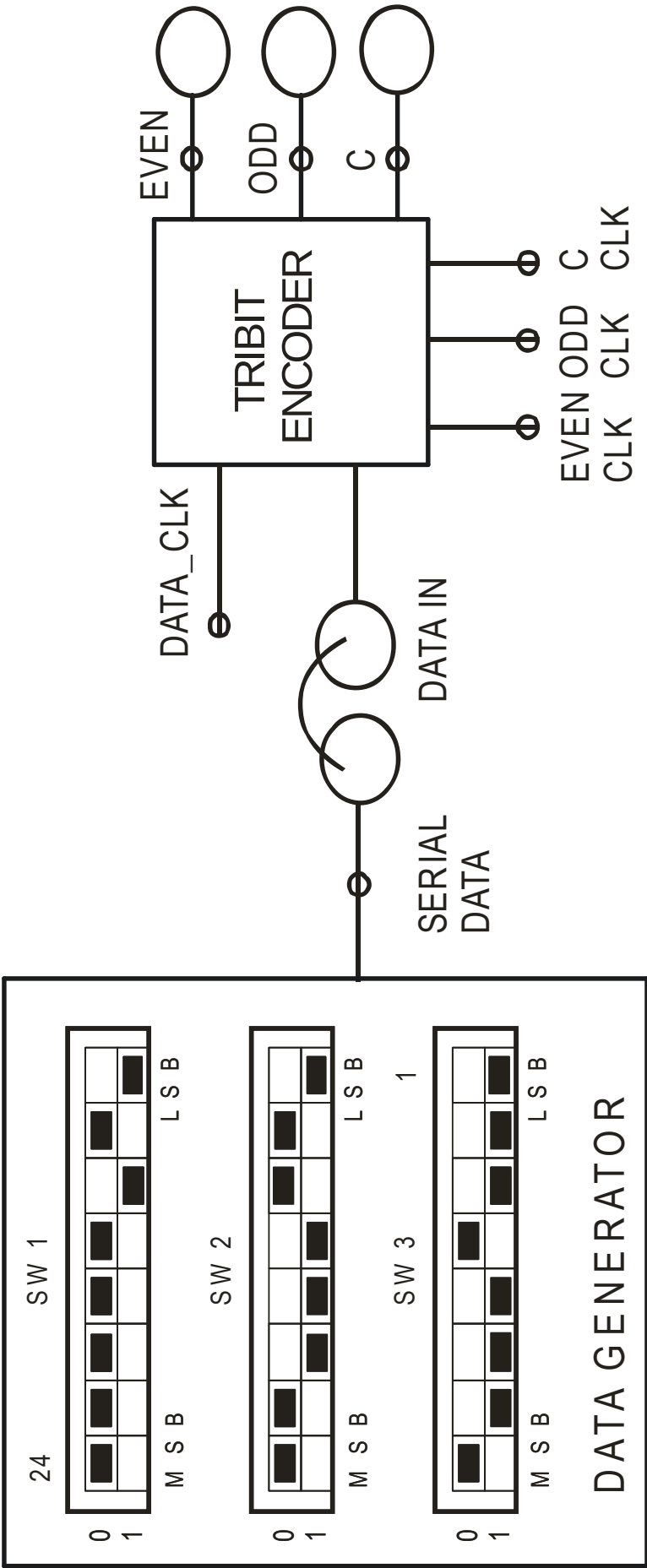


FIG 1.1 BLOCK DIAGRAM FOR TRIBIT ENCODING OF NRZ-L DATA

EXPERIMENT NO: 1

NAME:

TRIBIT DATA CODING TECHNIQUE OF NRZ-L DATA FORMAT

OBJECTIVE:

To study tribit coding techniques for Non-Return to Zero data format

THEORY:

NON - RETURN TO ZERO signal are the easiest formats that can be generated. These signals do not return to zero with the clock. The frequency component associated with these signals are half that of the clock frequency. Non-return to zero encoding is commonly used in slow speed communications interfaces for both synchronous and asynchronous transmission. Using NRZ, logic '1' bit is sent as a high value and logic '0' bit is sent as a low value.

The mechanism by which a bit stream $b(t)$ generates a QAM signal for transmission is shown in fig. 1.2 and relevant waveforms are shown in fig.1.3. In these waveforms we have arbitrarily assumed that in every case the active edge of the clock waveform is the upward edge. The D flip-flop generates three clocks waveforms. These clocks have period $3T_b$. The active edge of one of the clocks and the active edges of the other are separated by the bit time T_b . The bit stream $b(t)$ is applied as the data input to type-D flip-flops, driven by the three different clock waveform. Each flip-flop registers bits during positive transition of its clock in the bit stream $b(t)$ and holds each such registered bit for three bit intervals that is for the time $3T_b$. In fig.1.3 we have numbered the bits in the stream $b(t)$. Note that the bit stream EVEN bit (which is the output of the flip-flop driven by the EVEN clock) registers bit 1 and holds that bit for time $3T_b$, then register bit 4 for time $3T_b$, then bit 7 for $3T_b$, etc. The next bit stream ODD bit holds, for times $3T_b$ each, the bits numbered 2, 5, 8, etc. similarly the C bit stream holds the data bit numbered 3, 6, 9 for the bit interval $3T_b$.

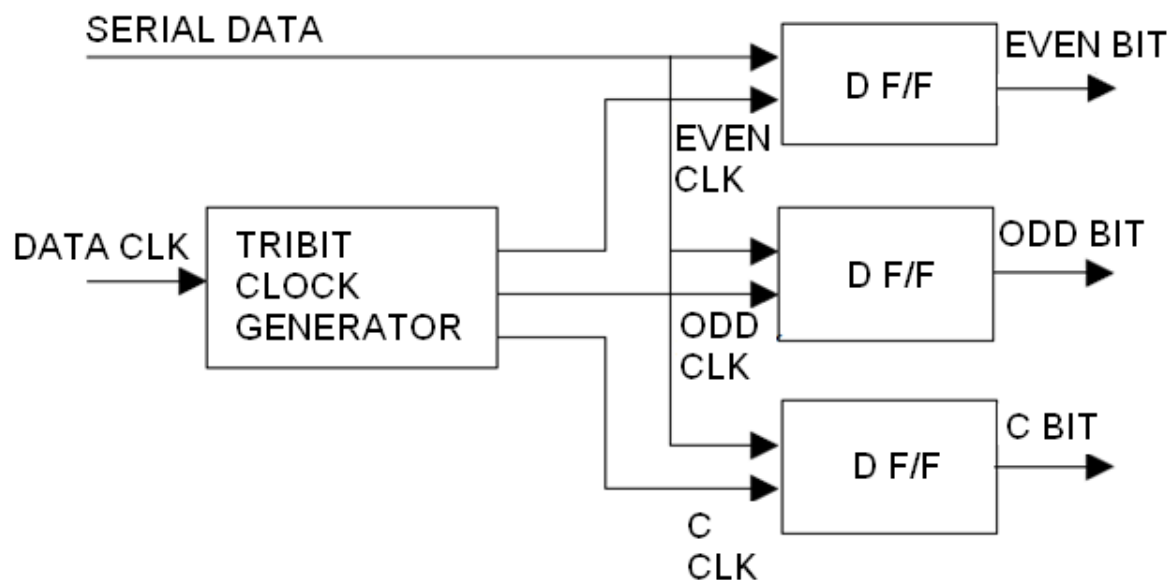
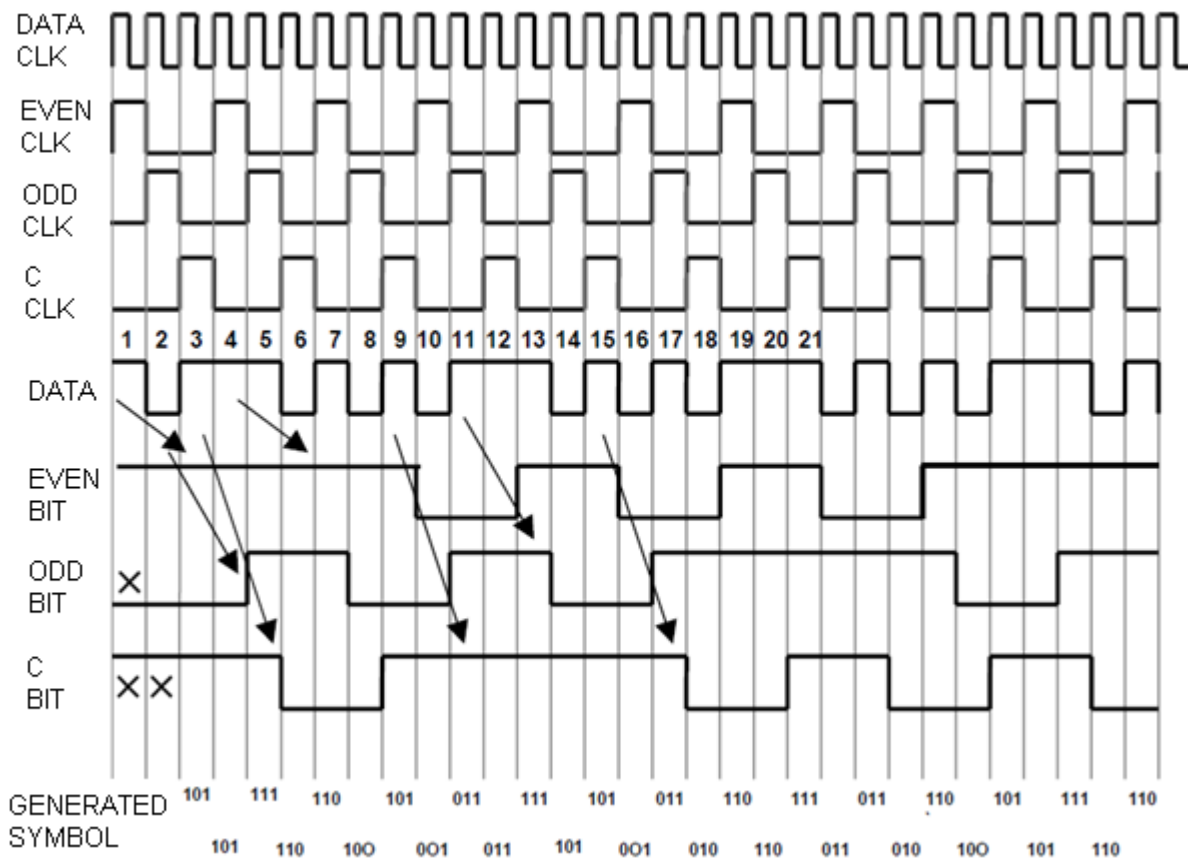


Fig. 1.2 TRIBIT CODING



EQUIPMENTS:

- DCL-QAM Kit
- Connecting Chords
- Power supply
- Dual Trace Oscilloscope

PROCEDURE:

1. Refer to the block diagram (Fig.1.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-QAM** and switch it on.
3. Select Data pattern of simulated data using switch **SW1, SW2, SW3**.
4. Connect **SERIAL DATA** generated to **DATA IN** of **TRIBIT ENCODER**.
5. Observe the tribit clock generated at **EVEN CLK**, **ODD CLK** and **C CLK** on the oscilloscope with respect to **DATA_CLOCK**. (Fig.1.4).
6. Observe the coded signal **EVEN**, **ODD** and **C** data on the oscilloscope with respect to **SERIAL DATA**. (Fig.1.4).

OBSERVATION:

Observe the following waveforms on oscilloscope and plot it on the paper.

1. Transmitter clock **DATA_CLOCK** with respect to NRZ-L coded Data **SERIAL DATA**. (Fig1.4 (a))
2. **EVEN CLK** with respect to **ODD CLK**. (Fig1.4 (b))
3. **EVEN CLK** with respect to **C CLK**. (Fig1.4 (c))
4. **EVEN** data with respect to **EVEN CLK**. (Fig1.4 (d))
5. **ODD** data with respect to **ODDCLK**. (Fig1.4 (e))
6. **C** data with respect to **CCLK**. (Fig1.4 (f))

CAPTURED WAVEFORM:

CH 1: DATA CLK (256 KHz) &

CH 2: SERIAL DATA (000001010011100101110111)

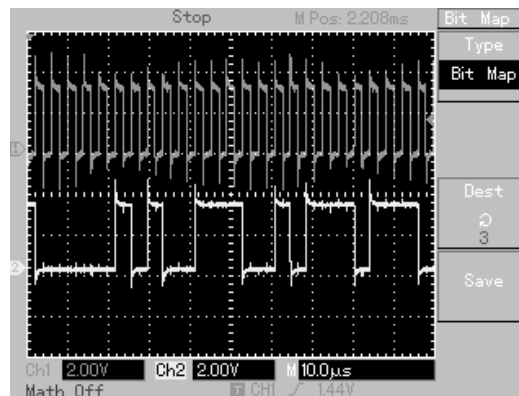


FIG. 1.4 (a)

CH 1: EVEN CLK & CH 2: ODD CLK

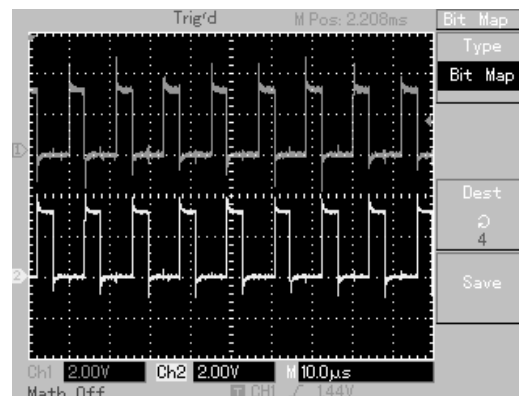
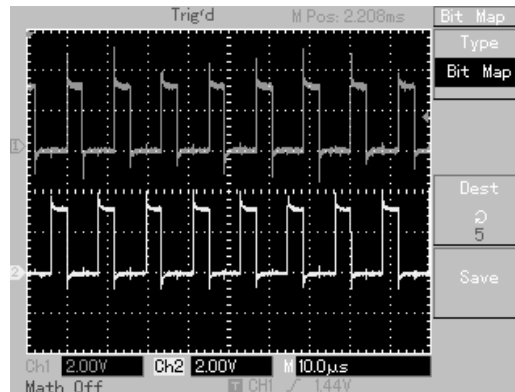
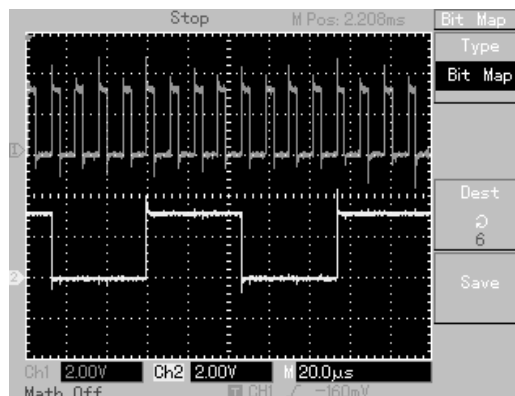
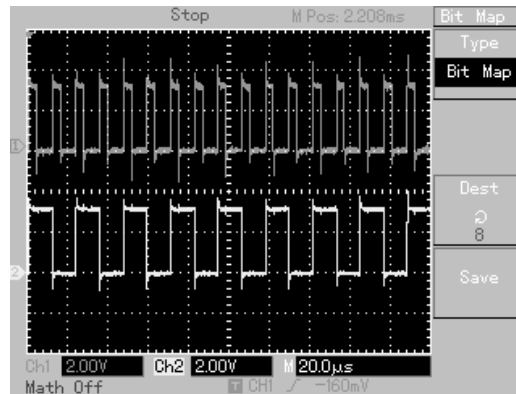


FIG. 1.4 (b)

CH 1: EVEN CLK & CH 2: C CLK**FIG. 1.4 (c)****CH 1: EVEN CLK & CH 2: EVEN (TRIBIT ENCODER)****FIG. 1.4 (d)****CH 1: ODD CLK & CH 2: ODD (TRIBIT ENCODER)****FIG. 1.4 (e)**

CH 1: C CLK & CH 2: C (TRIBIT ENCODER)**FIG. 1.4 (f)****CONCLUSION:**

The data are coded in Tribit, which generates, A data signal EVEN(in phase) consisting in voltage levels corresponding to the value of the first bit of the considered data, for a period equal to 3 bit intervals. A data signal ODD (in quadrature) consisting in voltage levels corresponding to the value of the second bit of the data, for duration equal to 3 bit intervals. A data signal C consisting in voltage levels corresponding to the value of the third bit of the data, for duration equal to 3 bit intervals.

EXPERIMENT NO.2

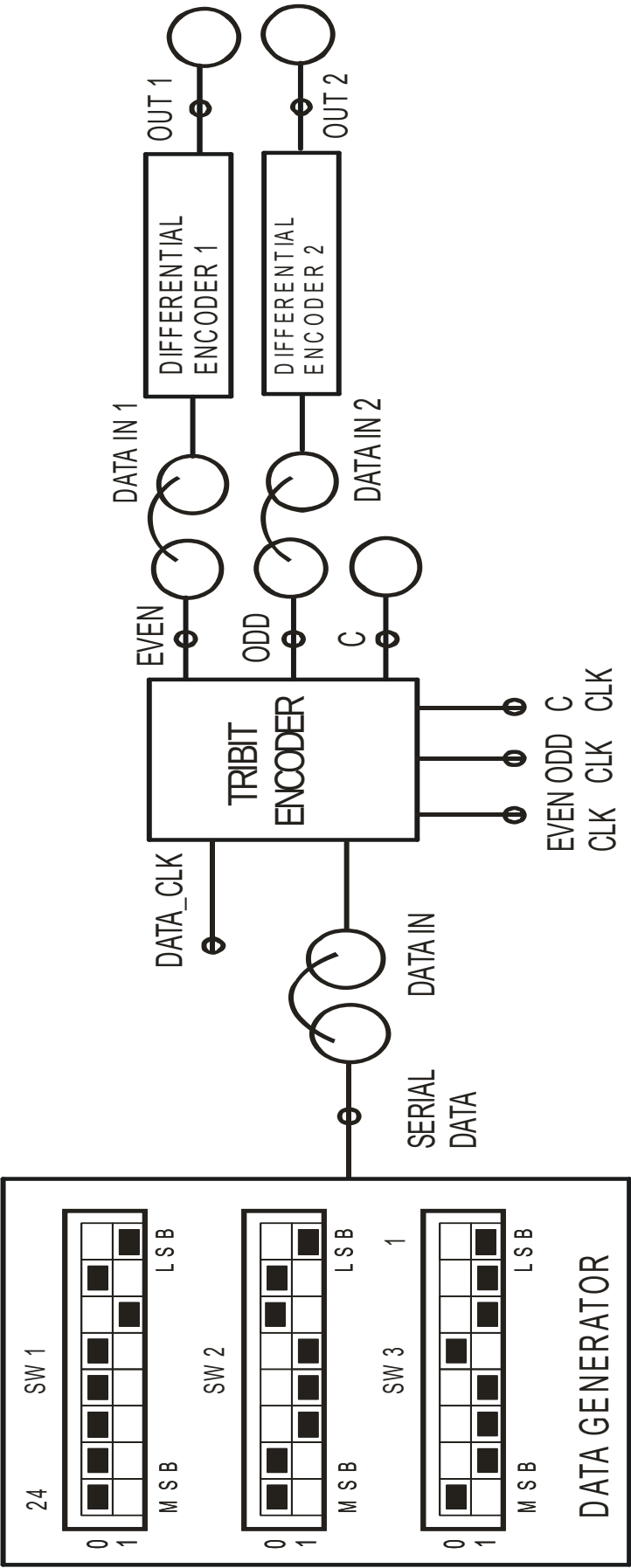


FIG 2.1 BLOCK DIAGRAM FOR DIFFERENTIAL ENCODING OF TRIBIT DATA

EXPERIMENT NO: 2

NAME:

DIFFERENTIAL ENCODING OF DATA

OBJECTIVE:

To study differential encoding techniques for data

THEORY:

Differential Encoding Logic:

A means of generating a differentially encoded data is shown in fig 2.2. The data stream to be transmitted, $d(t)$, is applied to one input of an Ex-OR logic gate. The other input of gate is the output of the Ex-OR gate $b(t)$ delayed by the time T_b allocated to one bit. This second input is then $b(t-T_b)$.

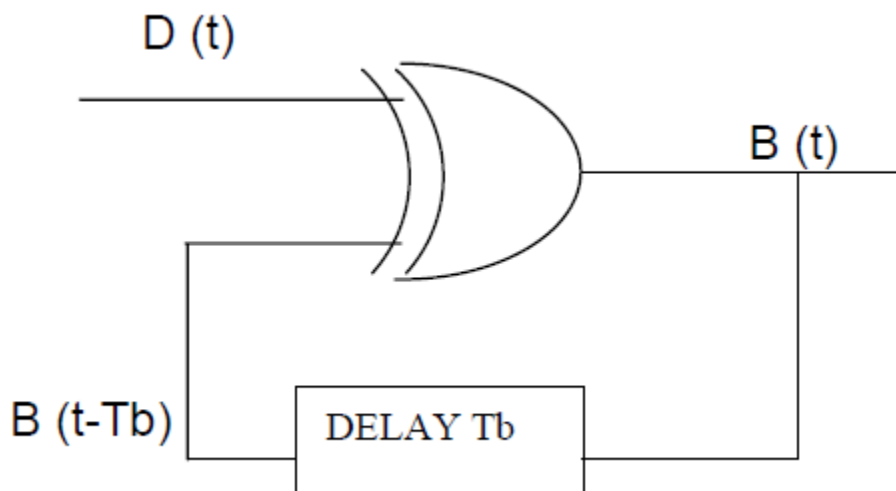


FIG 2.2 DIFFERENTIAL ENCODER

In fig 2.3 we have drawn the logic waveform to illustrate the response $b(t)$ to an input $d(t)$. The upper level of the waveform corresponds to logic 1, the lower level to logic 0. Because of the feedback involved in the system of fig 2.2 there is difficulty in determining the logic levels in the interval in which we start to draw the Waveform (interval 1 in fig 3.3) we cannot determine $b(t)$ in this first interval of our waveform unless we know $b(k=0)$. But we cannot determine $b(0)$ unless we know both $d(0)$ and $b(-1)$, etc. thus; to justify any set of logic levels in an initial bit interval we need to know the logic levels in the preceding interval. But such a determination requires information about the interval two bit times earlier and so on. In the waveform of fig 3.3 we have circumvented the problem by arbitrarily assuming that in the first interval $b(0) = 0$. It is shown below that, the data will be correctly determined regardless of our assumption concerning $b(0)$.

We now observe that the response of $b(t)$ to $d(t)$ is that $b(t)$ changes level at the beginning of each interval in which $d(t) = 1$ and $b(t)$ does not change level when $d(t) = 0$. Thus during interval 3, $d(3) = 1$, and correspondingly $b(3)$ changes at the beginning at that interval.

During interval 6 & 7, $d(6) = d(7) = 1$ and there are changes in $b(t)$ at the beginning of both intervals. During bits 10, 11, 12, and 13 $d(t) = 1$ and there are changes in $b(t)$ at the beginnings of each of these intervals. This behavior is to be anticipated from the truth tables of the exclusive-OR gate. For we note that when $d(t) = 0$, $b(t) = b(t-T_b)$ so that, whatever the initial value of DELAY T_b $b(t-T_b)$, it reproduces itself. On the other hand when $d(t) = 1$ then $b(t) = 1/b(t-T_b)$. Thus, in each successive bit intervals $b(t)$ changes from its value in the previous interval.

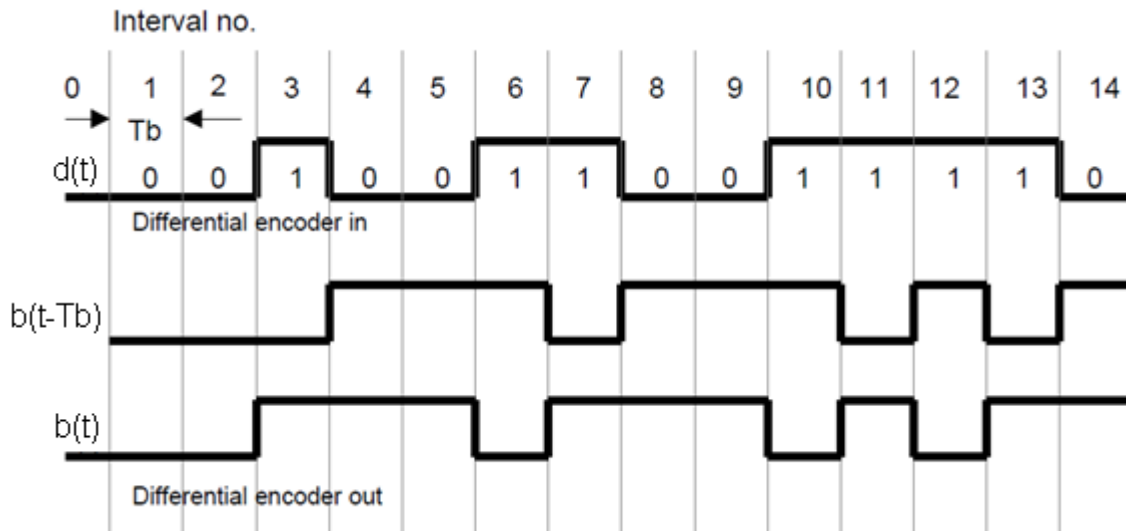


FIG 2.3 waveform of DIFFERENTIAL ENCODING of data

Note that in some intervals where $d(t) = 0$ we have $b(t) = 0$ and in other intervals when $d(t) = 0$ we have $b(t) = 1$. Similarly, when $d(t) = 1$ sometimes $b(t) = 1$ and sometimes $b(t) = 0$. Thus there is no correspondence between the levels of $d(t)$ and $b(t)$, and the only invariant feature of the system is that a change (some time up and sometime down) in $b(t)$ occurs whenever $d(t) = 1$, and that no change in $b(t)$ will occur whenever $d(t) = 0$.

Finally, we note that in waveform of fig 2.3 are drawn on the assumption that, in interval 1, $b(0) = 0$. As is easily verified, if not intuitively apparent, if we had assumed $b(0) = 1$, the invariant feature by which we have characterized the system would continue to apply. Since $b(0)$ must be either $b(0) = 0$ or $b(0) = 1$, there being no other possibilities, our result is valid quite generally. If, however, we had started with $b(0) = 1$ the levels $b(1)$ and $b(0)$ would have been inverted.

EQUIPMENTS:

- DCL-QAM Kit
- Connecting Chords.
- Power supply.
- Dual Trace Oscilloscope.

PROCEDURE:

1. Refer to the block diagram (Fig 2.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-QAM** and switch it on.
3. Select Data pattern of simulated data using switch **SW1, SW2, SW3,**
4. Connect **SERIAL DATA** generated to **DATA IN** of **TRIBIT ENCODER**.
5. Connect generated **EVEN** and **ODD** data to **DATA IN1** and **DATA IN2** of **DIFFERENTIAL ENCODERS** respectively.
6. Observe the input data and the differentially encoded **EVEN & ODD** data at **OUT1** and **OUT2** of **DIFFERENTIAL ENCODERS** respectively.

OBSERVATION:

ON KIT **DCL-QAM**

Observe the following waveforms on oscilloscope and plot it on the paper.

1. Differentially encoded data **OUT1** with respect to coded data **EVEN** (Fig.2.4 (a))
2. Differentially encoded data **OUT2** with respect to coded data **ODD** (Fig.2.4 (b))

CAPTURED WAVEFORM:

CH 1: EVEN (TRIBIT ENCODER) & CH 2: OUT1

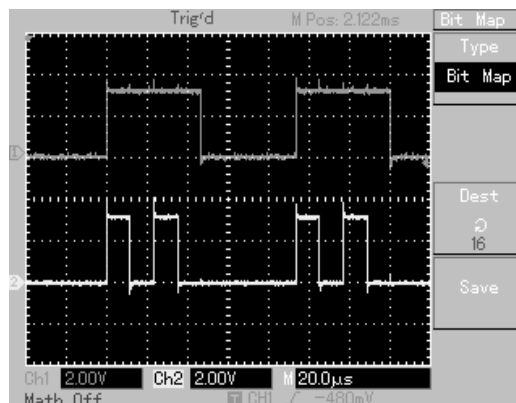


Fig.2.4 (a)

CH 1: ODD (TRIBIT ENCODER) & CH 2: OUT2

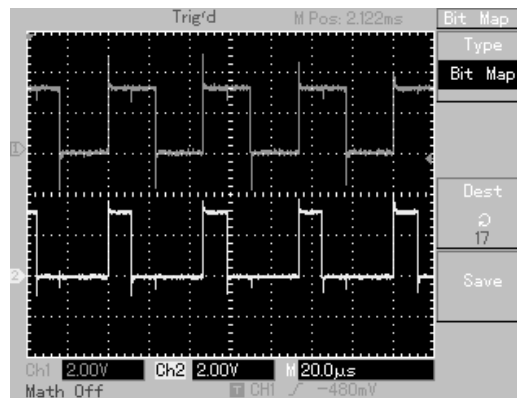


Fig.2.4 (b)

CONCLUSION:

It is observed that when input data to the differential encoder contains “1” the output of the differential encoder makes transition from its previous state. If input data contains “0” then output of differential encoder maintains the previous state.

EXPERIMENT NO.3

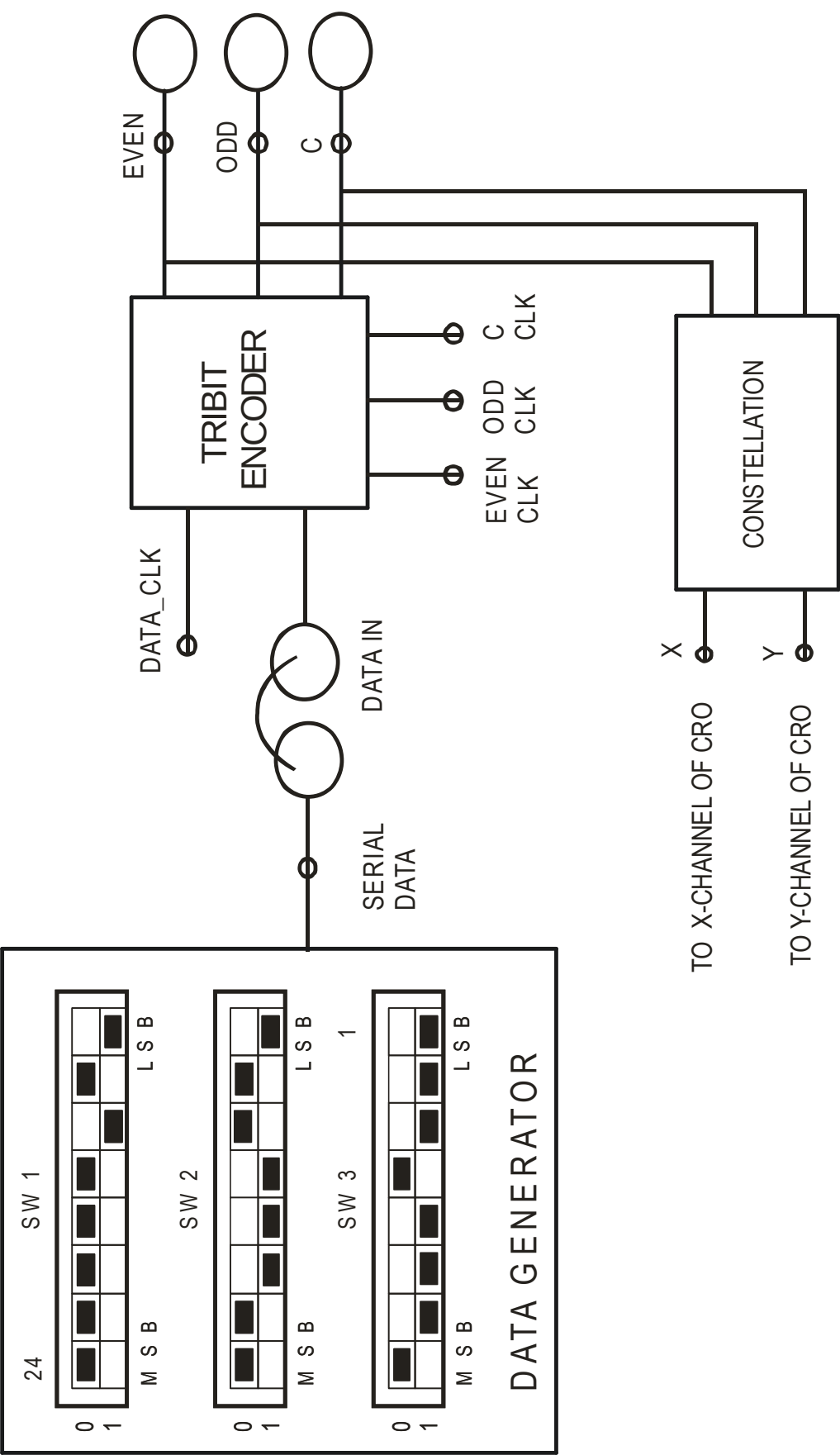


FIG 3.1 BLOCK DIAGRAM FOR OBSERVATION OF CONSTELLATION DIAGRAM

EXPERIMENT NO: 3

NAME:

OBSERVATION OF THE CONSTELLATION DIAGRAM FOR QAM AND STUDY OF BANDWIDTH EFFICIENCY OF QAM

OBJECTIVE:

- (A) To study the constellation diagram of QAM.
- (B) Study of bandwidth efficiency in Quadrature Amplitude Modulation techniques.

THEORY:

The constellation diagram or geometrical representation is as shown in fig 3.2 below. The point in signal space corresponding to each of the eight possible transmitted signals is indicated by dots. For each such signal we can recover three bits rather than one. The distance of a signal point from the origin is $\sqrt{E_s}$ which is the square root of the signal energy associated with the symbol, that is $E_s = P_s \cdot T_s = P_s(2T_b)$. As we know the ability to determine a bit without error is measured by the distance in signal space between points corresponding to the different values of the bit. We note from fig. that points which differ in a single bit are separated by the distance $d = 2\sqrt{P_s T_b} = 2\sqrt{E_b}$ where E_b is the energy contained in a bit transmitted for a time T_b . This distance for QPSK is the same as for BPSK. Hence, altogether, we have the important result that, in spite of the reduction by a factor of two in the bandwidth required by QPSK in comparison with BPSK, the noise immunity of the two systems are the same.

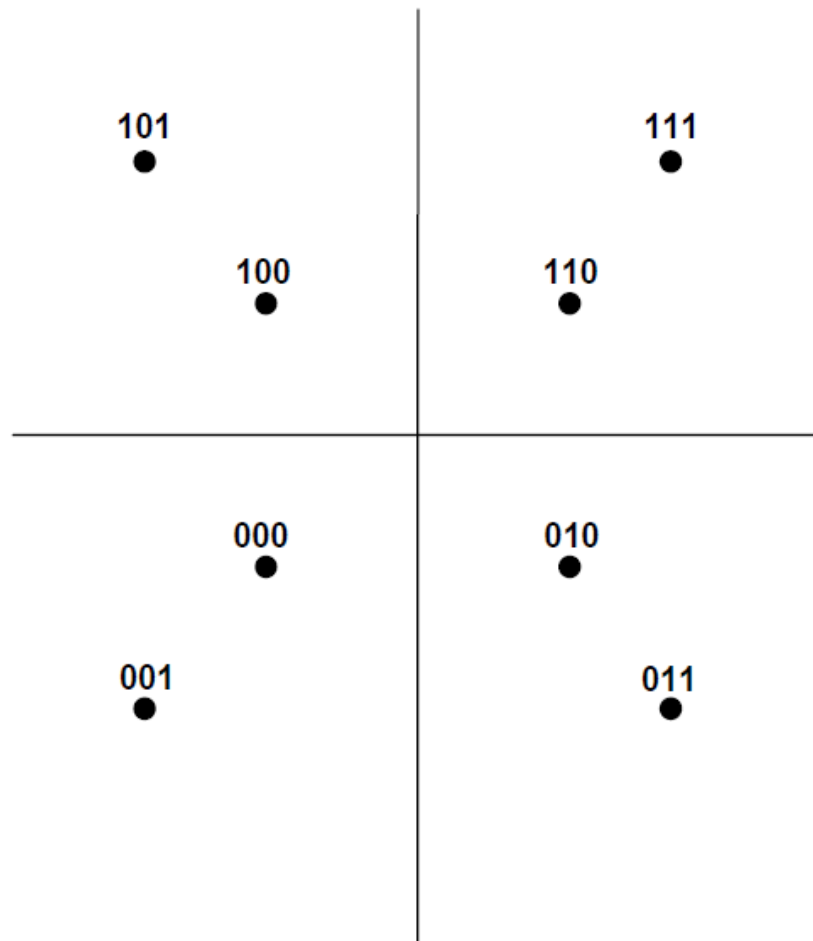


FIG 3.2 CONSTELLATION DIAGRAM

The number of dot points appearing in the constellation diagram depends on the number of symbol generated due to EVEN, ODD and C bit. The position of dot points in the quadrant of the constellation diagram is also depends on the symbol Generated due to the EVEN, ODD and C bit as shown above. The fig below shows examples of constellation diagram.

In fig 3.3 the data pattern used is 111001010011100101110000. If we plot the expected data pattern of EVEN, ODD and C bit as shown in fig, then there are 8symbols generated as 000, 001, 010, 011, 100, 101, 110, 111 thus there are 8dots appearing in the constellation diagram in all four quadrants.

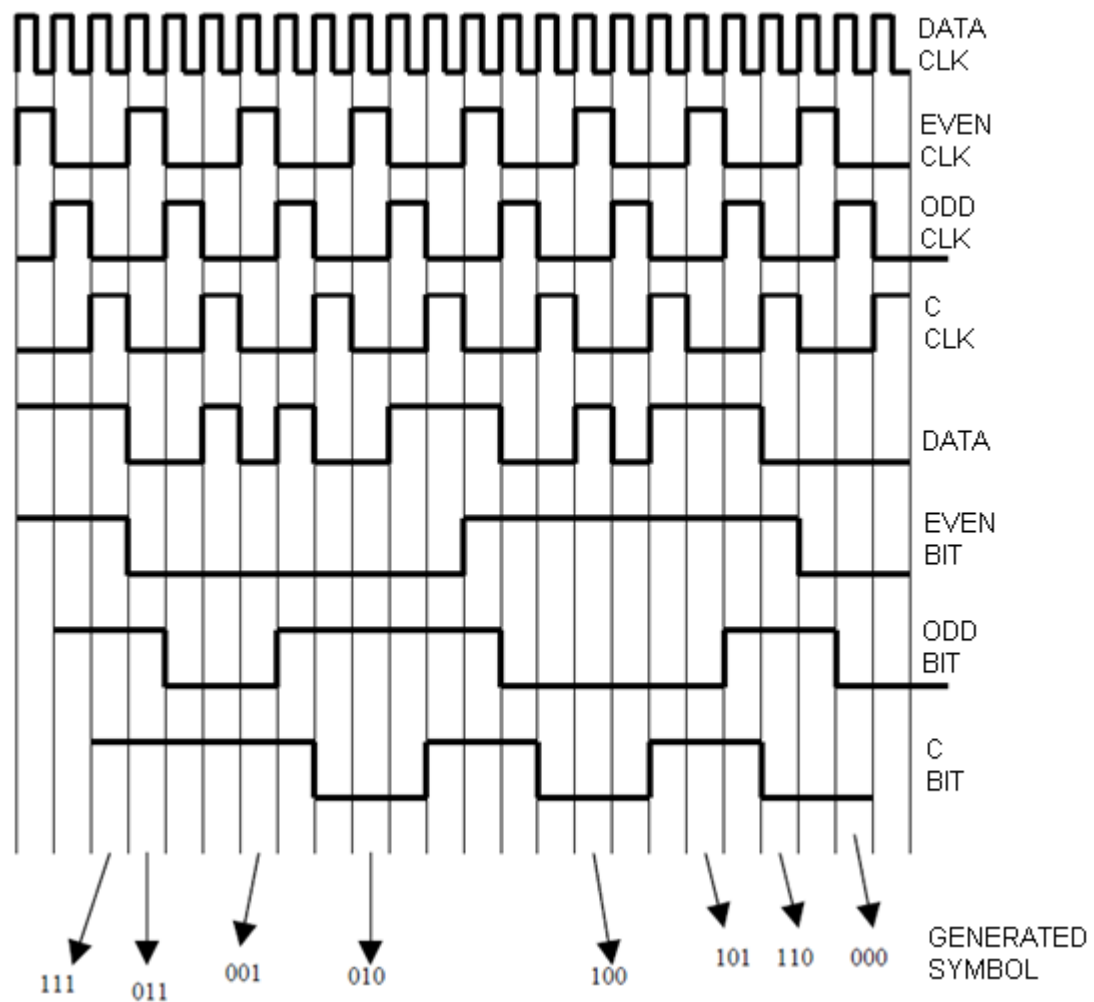


Fig 3.3

Bandwidth efficiency of QAM is defined by the ratio of bit transmission speed in QAM to Bandwidth of QAM signal. It is given by

$$\text{Bandwidth Efficiency} = F_b / B_w$$

Where F_b is Bit transmission speed and B_w is the bandwidth of signal transmitted.

The more the bandwidth efficiency more is the data transfer within the same bandwidth of signal transmission.

EQUIPMENTS:

- DCL-QAM Kit
- Connecting Chords.
- Power supply.
- Dual Trace Oscilloscope.

PROCEDURE:

(A) Study of constellation diagram.

1. Refer to the block diagram (Fig 3.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-QAM** and switch it on.
3. Select Data pattern of simulated data using switch **SW1, SW2, SW3**.
4. Connect **SERIAL DATA** generated to **DATA IN** of the **TRIBIT ENCODER**.
5. Now connect **X & Y** port of **CONSTELLATION BLOCK** to X-channel and Y-channel of CRO respectively
6. Observe waveforms as mentioned below (Fig 3. 4.).

OBSERVATION:

Observe the following waveforms on oscilloscope and plot it on the paper.
ON KIT **DCL-QAM**

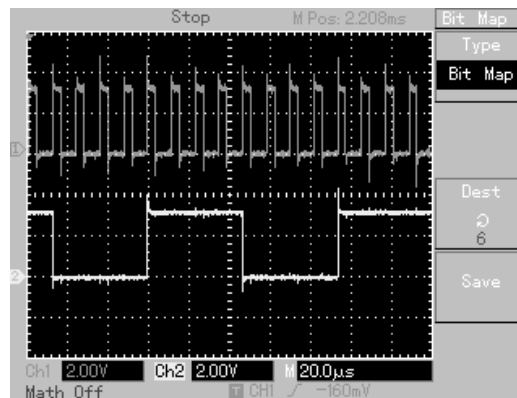
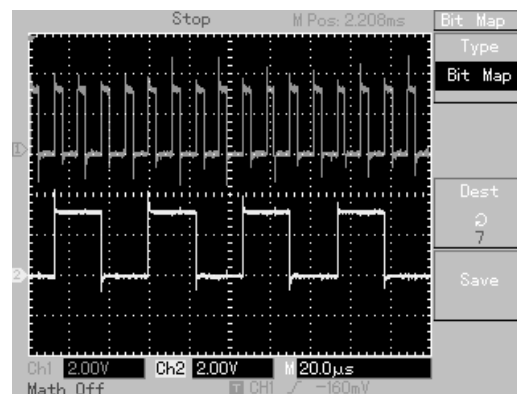
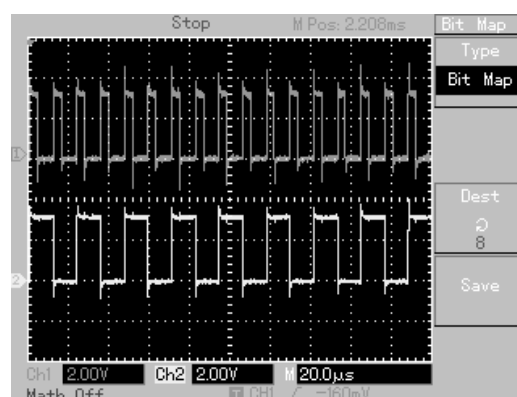
1. **SERIAL DATA** with respect to **DATA_CLK**. (Fig3.4 (a))
2. **EVEN** data with respect to **EVEN CLK**. (Fig3.4(b))
3. **ODD** data with respect to **ODDCLK**. (Fig3.4(c))
4. **C** data with respect to **CCLK**. (Fig3.4(d))
5. Constellation diagram on CRO for data pattern
6. "000001010011100101110111". (Fig3.4 (e))
7. Constellation diagram on CRO for data pattern
"101010101010101010101010". (Fig3.4 (f))

CAPTURED WAVEFORM:

**CH 1: DATA CLK (256 KHz) &
CH 2: SERIAL DATA (000001010011100101110111)**



FIG. 3.4 (a)

CH 1: EVEN CLK & CH 2: EVEN (TRIBIT ENCODER)**Fig.3.4 (b)****CH 1: ODD CLK & CH 2: ODD (TRIBIT ENCODER)****Fig.3.4 (c)****CH 1: C CLK & CH 2: C (TRIBIT ENCODER)****Fig.3.4 (d)**

SERIAL DATA: 000001010011100101110111
CH 1: X & CH 2: Y

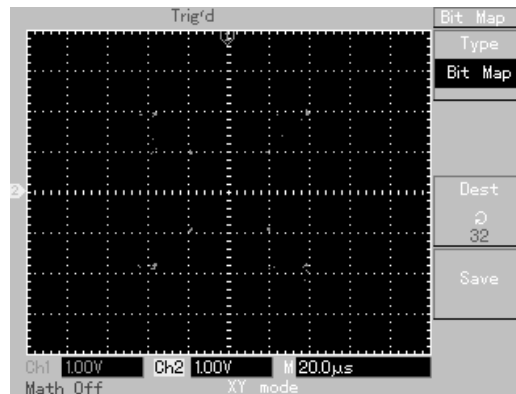


Fig. 3.4 (e)

SERIAL DATA: 101010101010101010101010
CH 1: X & CH 2: Y

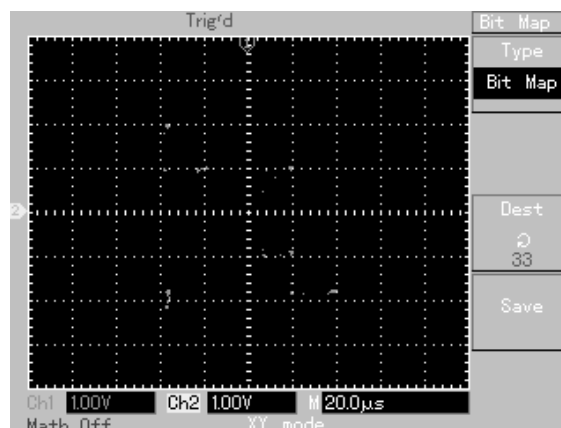


Fig. 3.4 (f)

(B) Bandwidth efficiency of QAM technique.

1. Measure the transmission clock at **DATA_CLOCK** post i.e. **256 KHz**
2. Divide this clock by the number of bits transmitted simultaneously, i.e. 3 Bits in this case (EVEN, ODD & C Bits).
3. Transmission clock is the bit transmission speed (256 KHz) and Bandwidth of modulator is clock divide by the number of bits transmitted (i.e. $256 \text{ KHz} / 3 = 85.33 \text{ KHz}$).
4. From the above data, calculate bandwidth efficiency by substituting these values in the formula as follows;

$$\begin{aligned} \text{Bandwidth Efficiency} &= F_b / B_w \\ \text{i.e.} \quad &= 256 / 85.33 \\ &= 3 \end{aligned}$$

CONCLUSION:

It is observed from the constellation diagram that, the symbol generated due to EVEN bit, ODD bit & C bit have eight possible values and lies in four quadrants. Also from the constellation diagram we can determine the distance between the two symbols. The transmitted signal, if it changes, changes phase by 90 deg. Rather than by 180 deg. as in BPSK.

Bandwidth Efficiency generally depends on the number of bits transmitted simultaneously

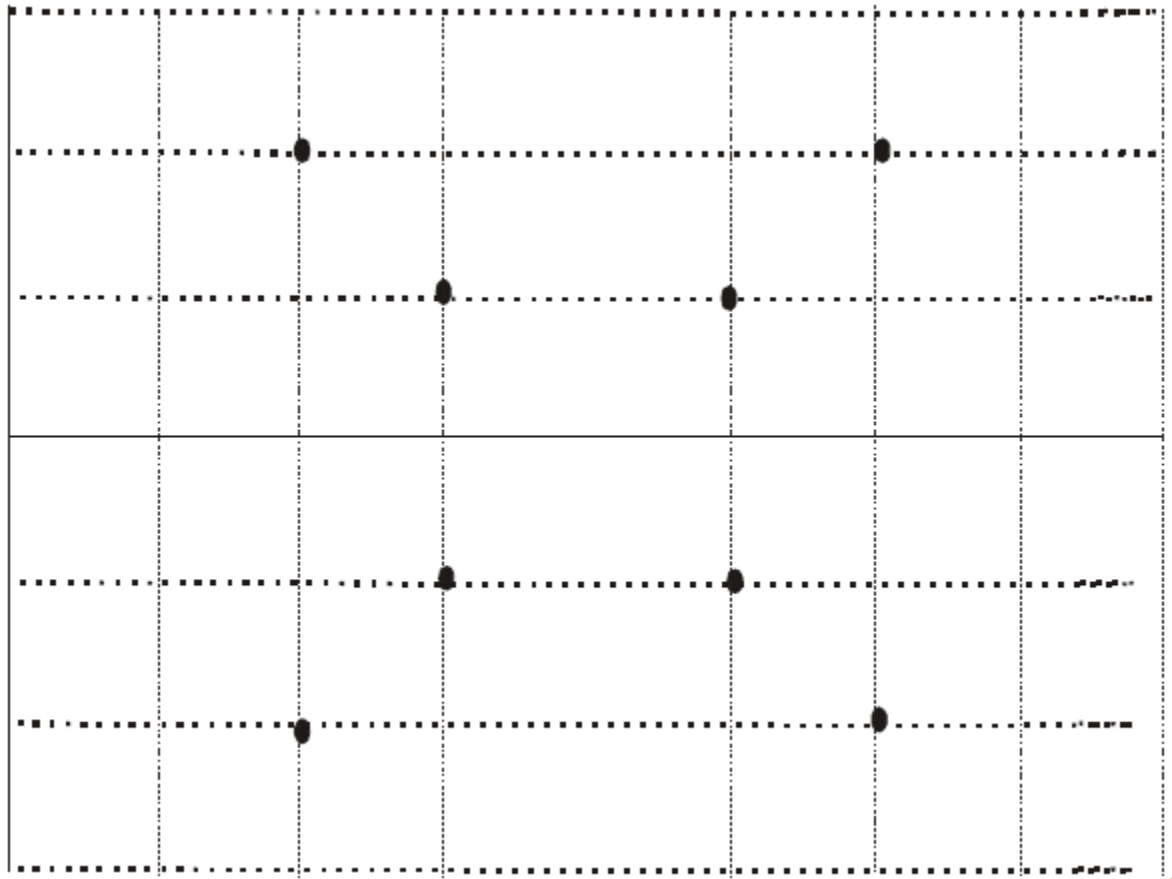


FIG. 3.4 CONSTELLATION DIAGRAM

EXPERIMENT NO.4

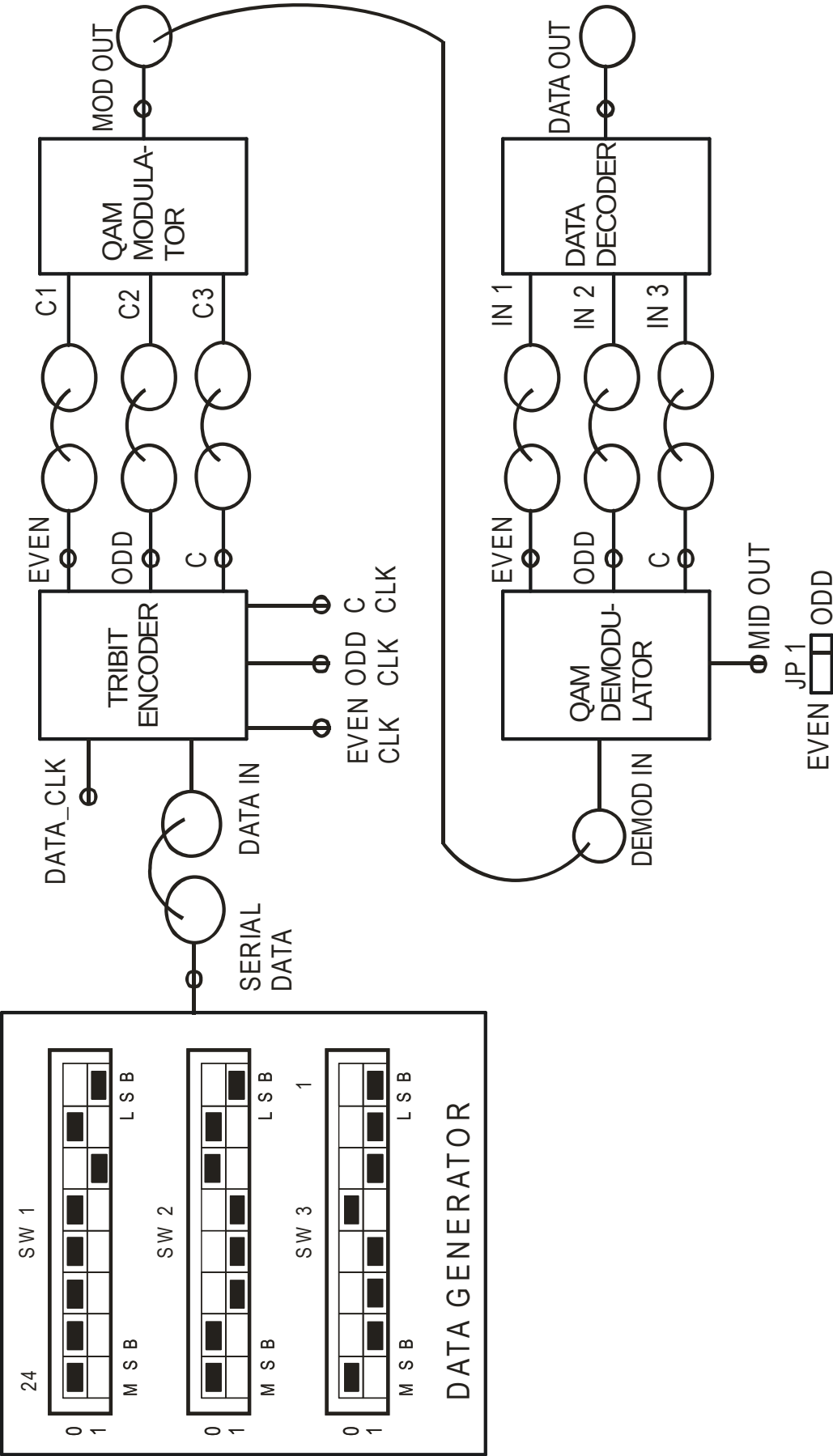


FIG 4.1 BLOCK DIAGRAM FOR QUADRATURE AMPLITUDE MODULATION TECHNIQUE

EXPERIMENT NO: 4

NAME:

QUADRATURE AMPLITUDE MODULATION TECHNIQUES

OBJECTIVE:

Study of Carrier Modulation Techniques by Quadrature Amplitude method

THEORY:

The QAM is a digital modulation where the information is contained into the phase as well as the amplitude of the transmitted carrier. In the 8-QAM the data are divided into the group of 3 bits (tribit), one of which will varies the amplitude of the carrier, the other two the phase. The modulated signal can take 4 different phases and 2 different amplitudes, for a total of 8 different states. Similarly in the 16-QAM the data are divided into group of 4 bits (quad bit). The 16 possible combinations change amplitude and phase of the carrier, which can take 16 different states.

A generator of 8-QAM signals for 3-bit symbol is shown in fig 4.2. For 8 QAM the main data source is divided into 3 bits called EVEN bit, ODD bit and C bit.

These three bits are called TRIBIT. These tribit together forms a symbol. We have 24 bit input data which gives 8 possible symbols. Among the 3 bits EVEN&ODD bit is responsible for phase modulation and the last bit (C bit) performs the amplitude modulation. The effect of each symbol on the final QAM signal is shown in the table below.

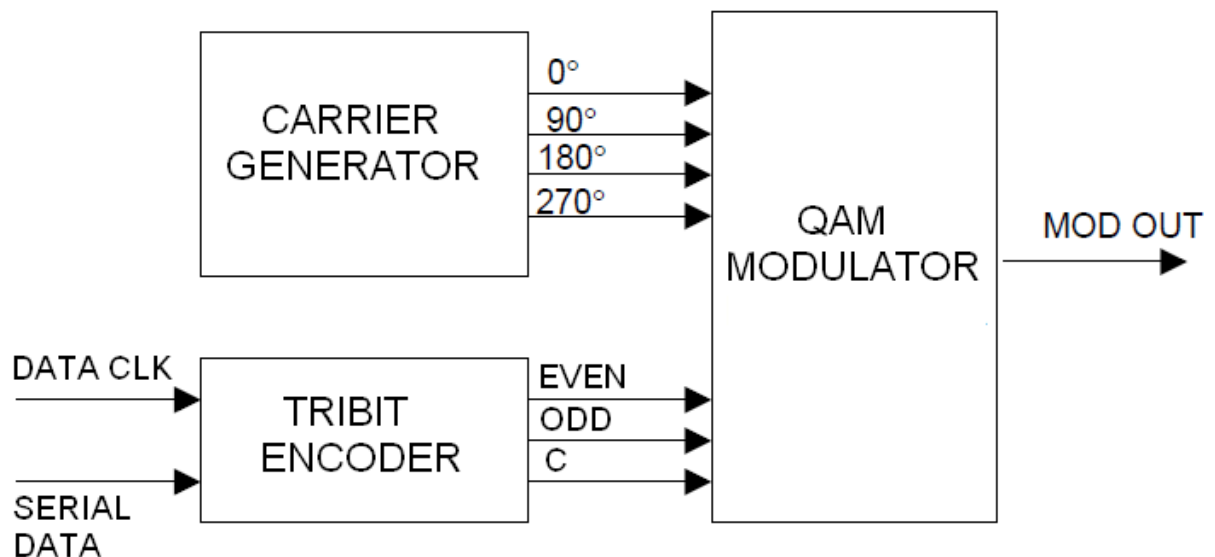


FIG 4.2 QAM MODULATOR

BINARY INPUT			QAM MOD OUT	
EVEN	ODD	C	AMPLITUDE	PHASE
0	0	0	1 V	180°
0	0	1	2 V	180°
0	1	0	1 V	90°
0	1	1	2 V	90°
1	0	0	1 V	270°
1	0	1	2 V	270°
1	1	0	1 V	0°
1	1	1	2 V	0°

The block diagram of the modulator used on the module is shown in the fig 4.1 four 1MHz sine carriers, shifted between them of 90 deg, are applied to modulator. The data (signal EVEN, ODD and C) reach the modulator from the Tribit coder. The instantaneous value of EVEN, ODD and C data bit generates a symbol. Since EVEN, ODD and C can take either 0 or 1 value, maximum 8 possible symbols can be generated as shown in the above table. According to the symbol generated one of the four-sine carriers will be selected. The relation between the symbol generated and sine carrier is shown in table above.

A receiver for the QAM signal is shown in fig 4.1 Synchronous detection is required and hence it is necessary to locally regenerate the carriers. The scheme for carrier regeneration is similar to BPSK. In that earlier case we used the multiplier and filter to recover the data.

The incoming signal is applied to the multipliers to remove the phase shift and to envelope detector for C bit recovery. The output of the multipliers can be seen at MID EVEN and MID OUT posts. The output of the multipliers is then given to filters, where we get the recovered EVEN and ODD data. These recovered EVEN&ODD bits having exactly same phase & frequency compared to transmitter EVEN&ODD bit. The C bit is recovered simply by passing the QAM modulated data through envelope detector. These recovered EVEN, ODD& C bits then applied to data decoder logic to recover the original NRZ-L data.

EQUIPMENTS:

- DCL-QAM Kit
- Connecting Chords.
- Power supply.
- Dual Trace Oscilloscope.

PROCEDURE:

1. Refer to the block diagram (Fig 4.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-QAM** and switch it on.
3. Select Data pattern of simulated data using switch **SW1, SW2, SW3**.
4. Connect **SERIAL DATA** generated to **DATA IN** of the **TRIBIT ENCODER**.
5. Connect the tribit data **EVEN, ODD & C** to control input **C1, C2** and **C3** of **QAM MODULATOR** respectively.
6. Connect QAM modulated signal **MOD OUT** of the **QAM MODULATOR** to the **DEMOD IN** of the **QAM DEMODULATOR**.
7. Observe output of multipliers at **MID OUT** post of **QAM DEMODULATOR** by changing jumper (JP1) at **EVEN** and **ODD** position respectively.
8. Connect the demodulated data **EVEN, ODD&C** of **QAM DEMODULATOR** to IN1, IN2, and IN3 of **DATA DECODER** respectively.
9. Observe the decoded data at **DATA OUT** post of **DATA DECODER**. Compare the decoded data with **SERIAL DATA**.
10. Observe various waveforms as mentioned below (Fig 4.3).

OBSERVATION:

Observe the following waveforms on oscilloscope and plot it on the paper.
ON KIT **DCL-QAM**

1. Transmitter clock **DATA_CLK** with respect to NRZ-L coded data **SERIAL DATA** (Fig4.3 (a))
2. **EVEN CLK** with respect to **ODD CLK**. (Fig4.3 (b))
3. **EVEN CLK** with respect to **C CLK**. (Fig4.3 (c))
4. **EVEN** data with respect to **EVEN CLK**. (Fig4.3(d))
5. **ODD** data with respect to **ODDCLK**. (Fig4.3 (e))
6. **C** data with respect to **CCLK**. (Fig4.3 (f))
7. Carrier signals **SIN 1, SIN 2** and **SIN 3** and **SIN 4**. (Fig4.3 (g,h))
8. **QAM** modulated signal at **MOD OUT** of **QAMMODULATOR** with respect to **EVEN** of **TRIBITENCODER**. (Fig4.3 (i))
9. **QAM** modulated signal at **MOD OUT** of **QAM MODULATOR** with respect to **C** of **TRIBIT ENCODER**. (Fig4.3 (j))
10. Mid **EVEN** signal at **MID OUT** post of **QAM DEMODULATOR** with respect to **EVEN** of **TRIBITENCODER**. (Fig4.3 (k))
11. Mid **ODD** signal at **MID OUT** post of **QAM DEMODULATOR** with respect to **ODD** of **TRIBIT ENCODER**. (Fig4.3 (l))
12. **EVEN** of **QAM DEMODULATOR** with respect to **EVEN** of **TRIBIT ENCODER**. (Fig4.3 (m))

13. **ODD of QAM DEMODULATOR** with respect to **ODD of TRIBIT ENCODER.** (Fig4.3 (n))
14. **C of QAM DEMODULATOR** with respect to **C of TRIBIT ENCODER.** (Fig4.3 (o))
15. **DATA OUT** with respect to **SERIAL DATA.** (Fig4.3 (p))

CAPTURED WAVEFORM:

**CH 1: DATA CLK (256 KHz) &
CH 2: SERIAL DATA (000001010011100101110111)**

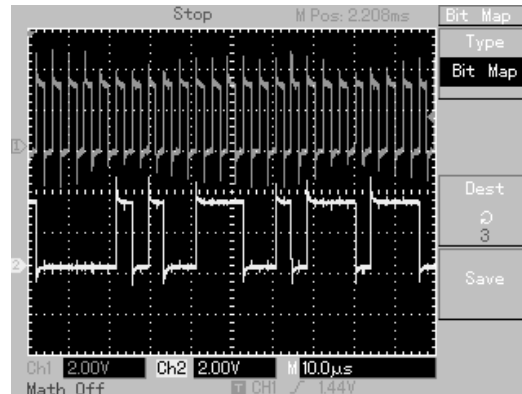
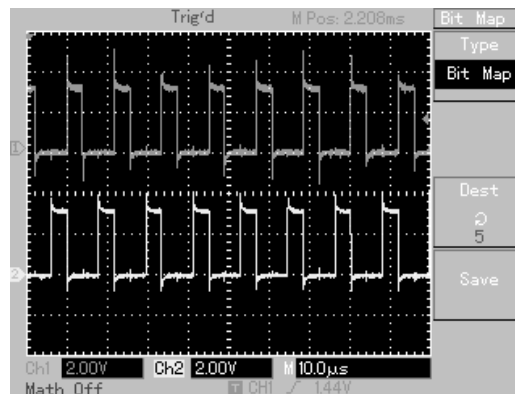
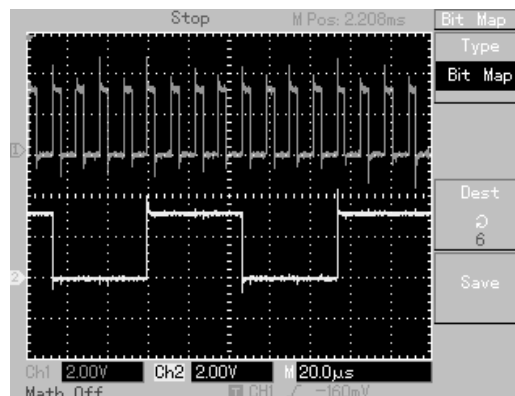


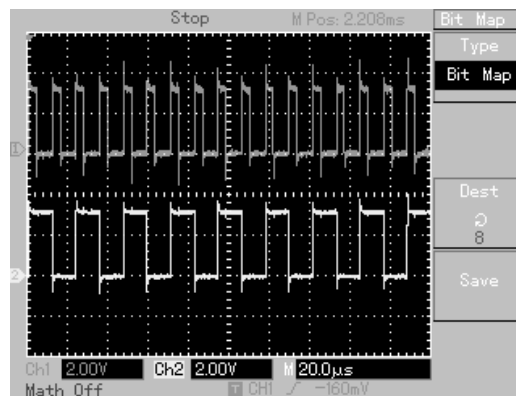
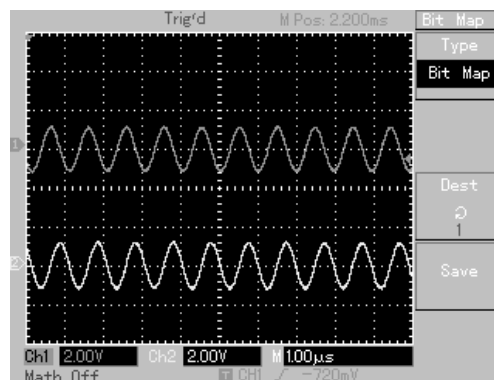
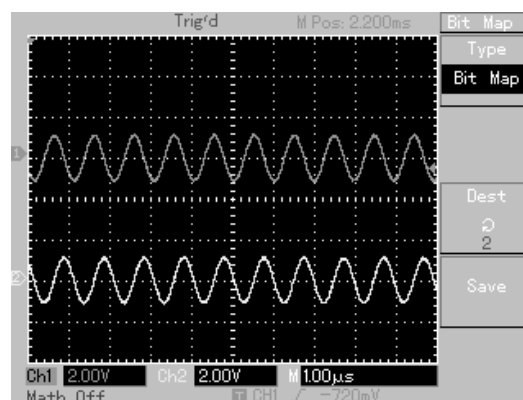
FIG. 4.3 (a)

CH 1: EVEN CLK & CH 2: ODD CLK



FIG. 4.3 (b)

CH 1: EVEN CLK & CH 2: C CLK**FIG. 4.3 (c)****CH 1: EVEN CLK & CH 2: EVEN (TRIBIT ENCODER)****FIG. 4.3 (d)****CH 1: ODD CLK & CH 2: ODD (TRIBIT ENCODER)****FIG. 4.3 (e)**

CH 1: C CLK & CH 2: C (TRIBIT ENCODER)**FIG. 4.3 (f)****CH 1: SIN 1 (0°) & CH 2: SIN 2 (90°)****Fig.4.3 (g)****CH 1: SIN 3 (180°) & CH 2: SIN 4 (270°)****Fig.4.3 (h)****NOTE:**

For **MOD OUT**, select a symbol using **EVEN** and **ODD** and then observe the **MOD OUT**. For waveform below a symbol is selected using **EVEN** and **ODD** and then **MOD OUT** is observed with respect to **EVEN**.

CH 1: EVEN (TRIBIT ENCODER) & CH 2: MOD OUT

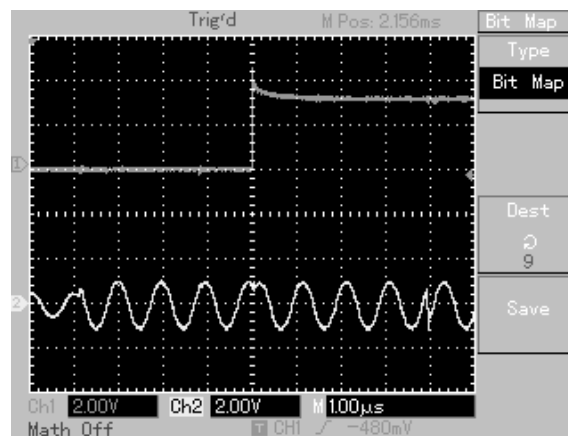


Fig.4.3 (i)

CH 1: C (TRIBIT ENCODER) & CH 2: MOD OUT

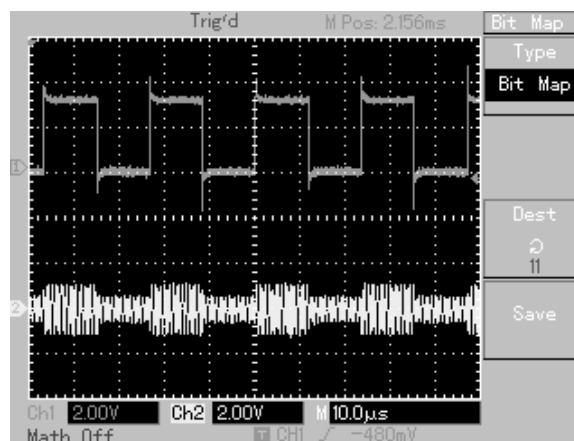


Fig.4.3 (j)

CH 1: EVEN (TRIBIT ENCODER) & CH 2: MID OUT

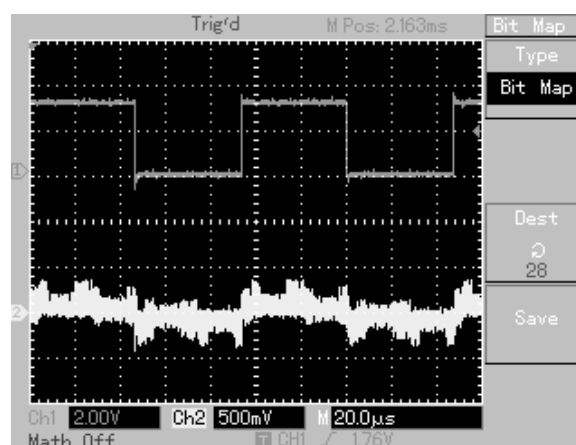
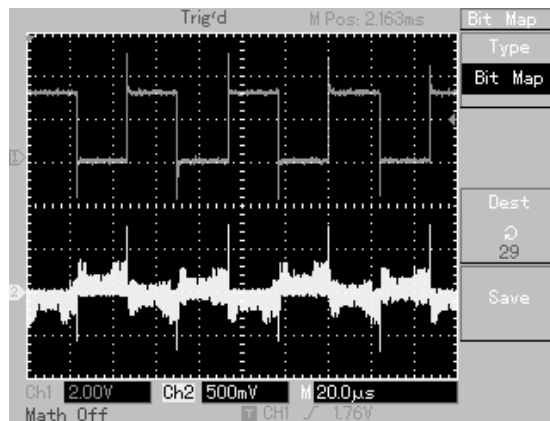
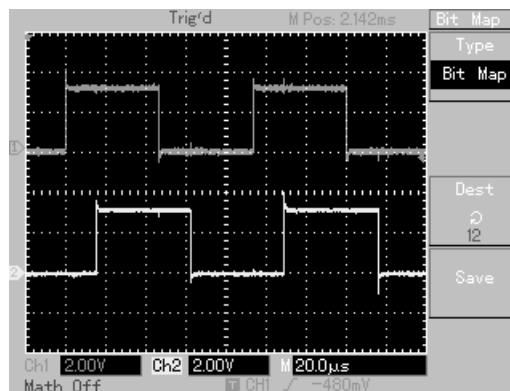
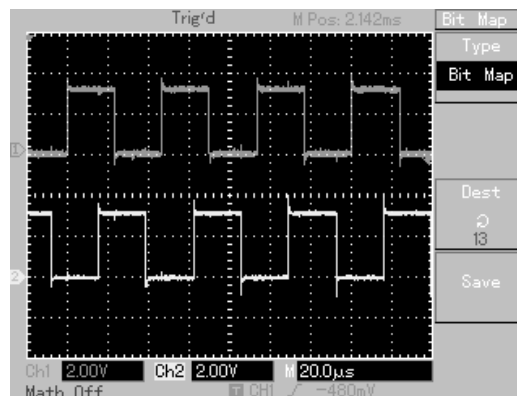
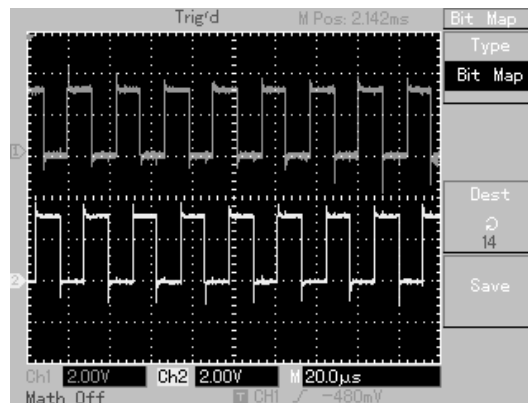
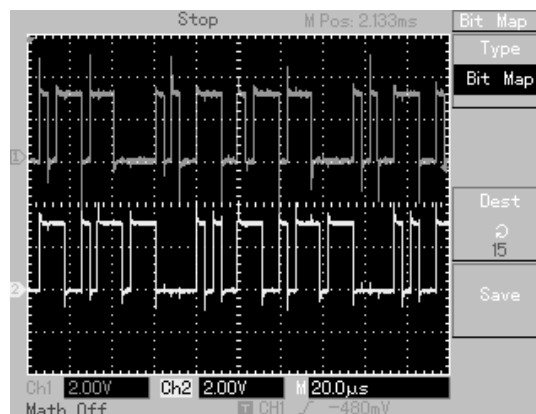


Fig.4.3 (k)

CH 1: ODD (TRIBIT ENCODER) & CH 2: MID OUT**Fig.4.3 (l)****CH 1: EVEN (TRIBIT ENCODER) & CH 2: EVEN (Demodulated)****Fig.4.3 (m)****CH 1: ODD (TRIBIT ENCODER) & CH 2: ODD (Demodulated)****Fig.4.3 (n)**

CH 1: C (TRIBIT ENCODER) & CH 2: C (Demodulated)**Fig.4.3 (o)****CH 1: SERIAL DATA & CH 2: DATA OUT****Fig.4.3 (p)****CONCLUSION:**

In BPSK we deal individually with each bit of duration T_b . In QAM we lump three bits together to form a SYMBOL. The symbol can have any one of eight possible values corresponding to three-bit sequence. We therefore arrange to make available for transmission eight distinct signals. At the receiver each signal represents one symbol and, correspondingly, three bits. When bits are transmitted, as in BPSK, the signal changes occur at the bit rate. When symbols are transmitted the changes occur at the symbol rate, which is one-third the bit rate. Thus the symbol time is $T_s = 3T_b$.

EXPERIMENT NO.5

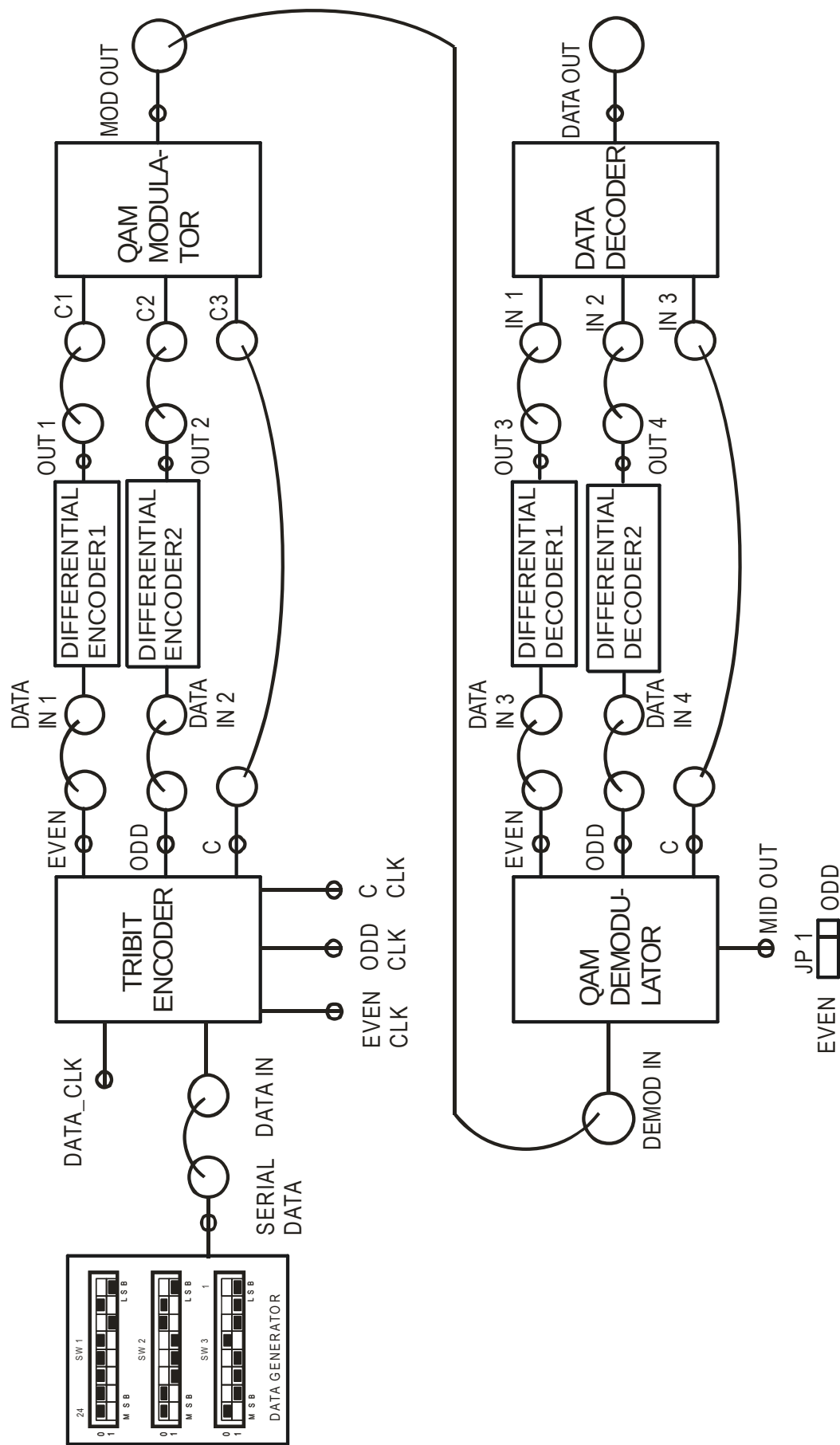


FIG 5.1 BLOCK DIAGRAM FOR STUDY OF DIFFERENTIAL QUADRATURE AMPLITUDE MODULATION TECHNIQUE

EXPERIMENT NO: 5

NAME:

DIFFERENTIAL QUADRATURE AMPLITUDE MODULATION TECHNIQUE

OBJECTIVE:

Study of carrier modulation technique by Differential Quadrature Amplitude modulation method

THEORY:

In DQAM or 8-differential QAM modulation, the phase of the sine carrier is changed as function of the pair of data bit (EVEN&ODD bit). The modulator is same used for QAM. Before entering the modulator the data are coded so to generate two” differential” signals namely DIFF_EVEN and DIFF_ODD. These consider the variation of the bit pairs, in respect to the pair just before. The block diagram of DQAM is shown below:

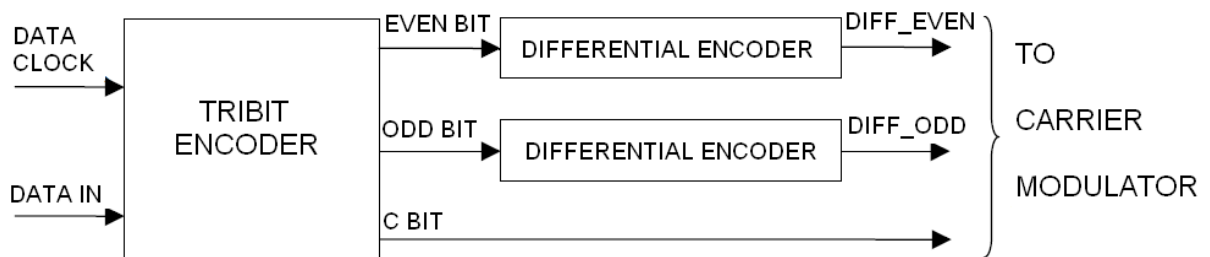


FIG 5.2A BASIC DQAM MODULATION.

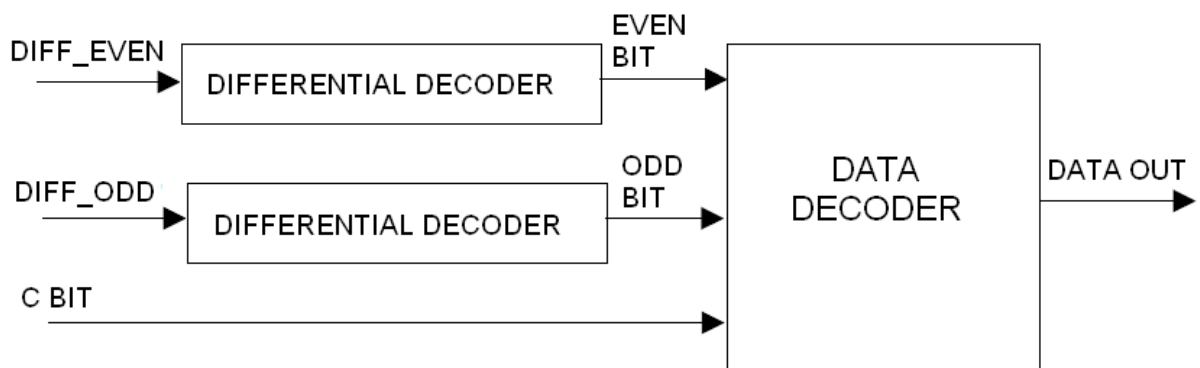


FIG 5.2B BASIC DQAM DEMODULATION

The differential coding of data to be transmitted makes the “bit pair” to be transformed into phase variation of the carrier. In this way the receiver recognizes the pair each time it detects a phase shift of the modulated carrier independently from its absolute phase. The ambiguity of the QPSK modulation, which can take to the demodulated data inversion, is solved in this way.

The procedure for generating the DQAM signal is same as QAM, but in DQAM the EVEN&ODD bits are differentially encoded using separate differential encoder as shown in fig 5.2A While C bit remains same. The concept of differential encoding is same as explained in experiment no 2. After differential encoding the encoded data is fed to control inputs of carrier modulator. The four phase shifted carrier is also fed to the carrier modulator. The output of the carrier modulator is DQAM modulated signal. The recovery of DQAM signal is same up to the EVEN, ODD& C bit recovery. In DQAM we first recover the differentially encoded EVEN & ODD bits. These recovered bits need to be decoded to bring it into original EVEN&ODD bit format. Thus, two separate differential decoders are used at the DQAM demodulator. The block diagram of differential decoder is shown in fig 5.3.

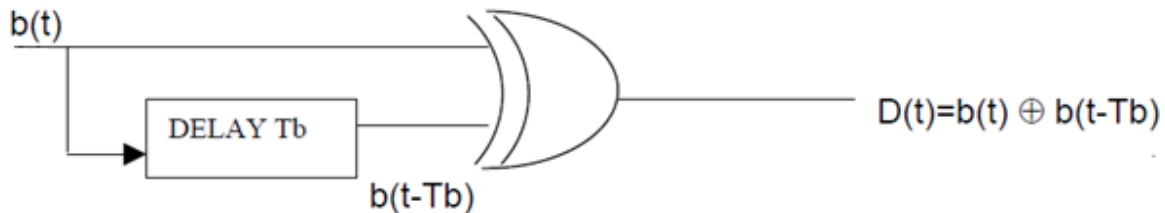


FIG 5.3 DIFFERENTIAL DECODER

The logic of differential decoder is, the recovered differentially encoded EVEN&ODD bits are applied directly to one input of an exclusive-OR logic gate and to the other input is applied $b(t-Tb)$ (see fig 5.3) i.e. one bit delayed. The gate output will be at one or the other of its level depending on whether $b(t)=b(t-Tb)$ or $b(t) \neq b(t-Tb)$. In the first case $b(t)$ did not change level and therefore the transmitted bit is $D(t)=0$. In second case $d(t)=1$.

At the output of differential decoder we get the original EVEN&ODD bits. These EVEN & ODD bits then applied to the data decoder along with decoder clock to recover the original NRZ-L data.

EQUIPMENTS:

- DCL-QAM Kit.
- Connecting Chords.
- Power supply.
- Dual Trace Oscilloscope.

PROCEDURE:

1. Refer to the block diagram (Fig 5.1) and carry out the following connections and switch settings.
2. Connect power supply in proper polarity to the kit **DCL-QAM** and switch it on.
3. Select Data pattern of simulated data using switch **SW1, SW2, SW3**.
4. Connect **SERIAL DATA** generated to **DATA IN** of the **TRIBIT ENCODER**.
5. Connect generated **EVEN** and **ODD** data to **DATA IN1** and **DATA IN2** of **DIFFERENTIAL ENCODERS** respectively.
6. Connect **OUT1** and **OUT2** to control input **C1** and **C2** of **QAM MODULATOR** respectively.
7. Connect **C** of **TRIBIT ENCODER** to control input **C3** of **QAMMODULATOR**.
8. Connect modulated signal **MOD OUT** to the **DEMOD IN** of the **QAM DEMODULATOR**.

9. Connect demodulated **EVEN** and **ODD** data of **QAM DEMODULATOR** to **DATA IN3** and **DATA IN4** of **DIFFERENTIAL DECODERS** respectively.
10. Connect **OUT3** and **OUT4** to input **IN1** and **IN2** of **DATA DECODER** respectively.
11. Connect **C** of **QAM DEMODULATOR** to **IN3** of **DATA DECODER**.
12. Observe the decoded data at **DATA OUT** post of **DATA DECODER** and Compare the decoded data with **SERIAL DATA**.
13. Observe various waveforms as mentioned below (Fig. 5.4).

OBSERVATION:

Observe the following waveforms on oscilloscope and plot it on the paper.

ON KIT **DCL-QAM**

1. Transmitter clock **DATA_CLK** with respect to NRZ-L coded data **SERIAL DATA**. (Fig5.4 (a))
2. **EVEN CLK** with respect to **ODD CLK**. (Fig5.4 (b))
3. **EVEN CLK** with respect to **C CLK**. (Fig5.4 (c))
4. **EVEN** data with respect to **EVEN CLK**. (Fig5.4 (d))
5. **ODD** data with respect to **ODDCLK**. (Fig5.4 (e))
6. **C** data with respect to **CCLK**. (Fig5.4 (f))
7. Differentially encoded data **OUT1** with respect to coded data **EVEN**. (Fig5.4 (g))
8. Differentially encoded data **OUT2** with respect to coded data **ODD**. (Fig5.4 (h))
9. Carrier signals **SIN 1**, **SIN 2** and **SIN 3** and **SIN 4**. (Fig5.4 (i,j))
10. **DQAM** modulated signal at **MOD OUT** of **QAM modulator** with respect to **OUT1**. (Fig5.4 (k))
11. Mid **EVEN** signal at **MID OUT** post of **QAM DEMODULATOR** with respect to **OUT1**. (Fig5.4 (l))
12. Mid **ODD** signal at **MID OUT** post of **QAM DEMODULATOR** with respect to **OUT2**. (Fig5.4 (m))
13. **EVEN** of **QAM DEMODULATOR** with respect to **OUT1**. (Fig5.4 (n))
14. **ODD** of **QAM DEMODULATOR** with respect to **OUT2**. (Fig5.4 (o))
15. **C** of **QAM DEMODULATOR** with respect to **C** of **TRIBIT ENCODER**. (Fig5.4 (p))
16. **OUT3** with respect to **EVEN** of **TRIBIT ENCODER**. (Fig5.4 (q))
17. **OUT4** with respect to **ODD** of **TRIBIT ENCODER**. (Fig5.4 (r))
18. **DATA OUT** with respect to **SERIAL DATA**. (Fig5.4 (s))

CAPTURED WAVEFORM:

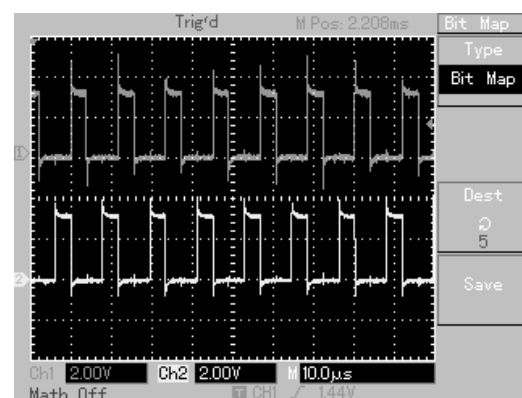
**CH 1: DATA CLK (256 KHz) &
CH 2: SERIAL DATA (000001010011100101110111)**

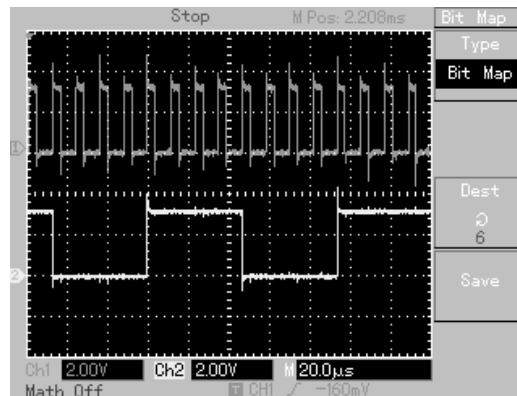
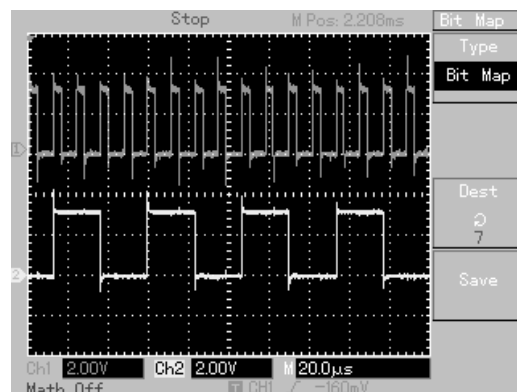
**FIG. 5.4 (a)**

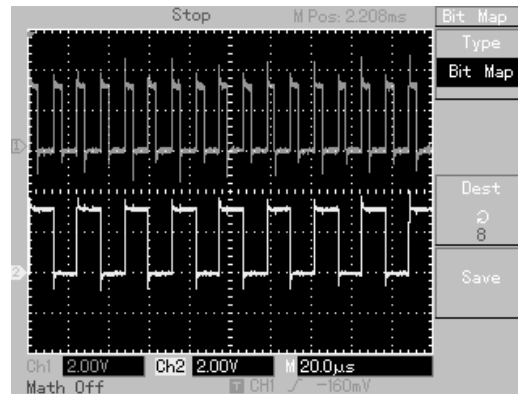
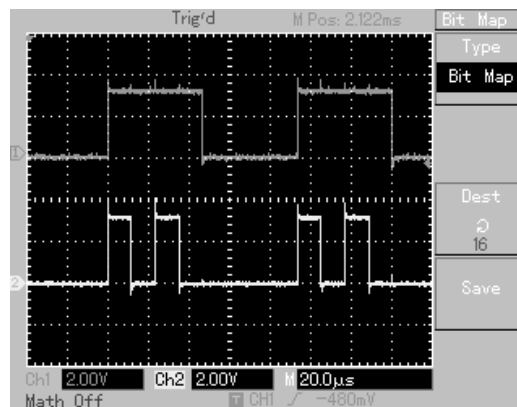
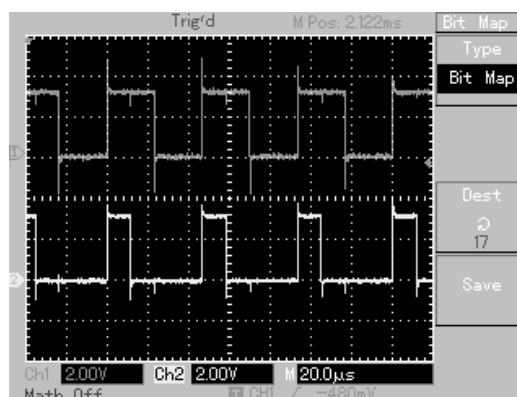
CH 1: EVEN CLK & CH 2: ODD CLK

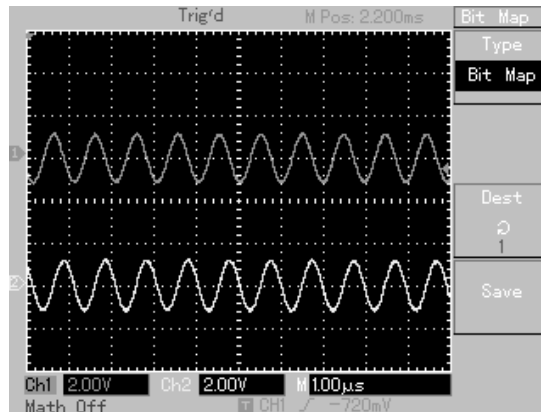
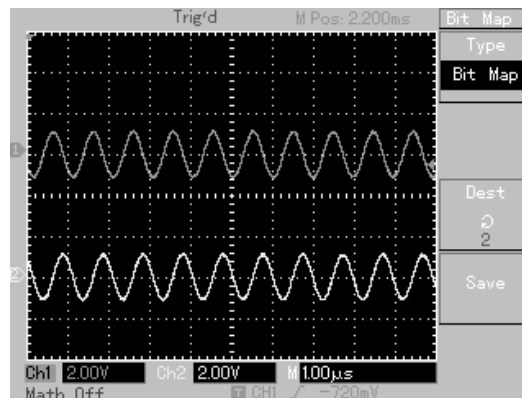
**FIG. 5.4 (b)**

CH 1: EVEN CLK & CH 2: C CLK

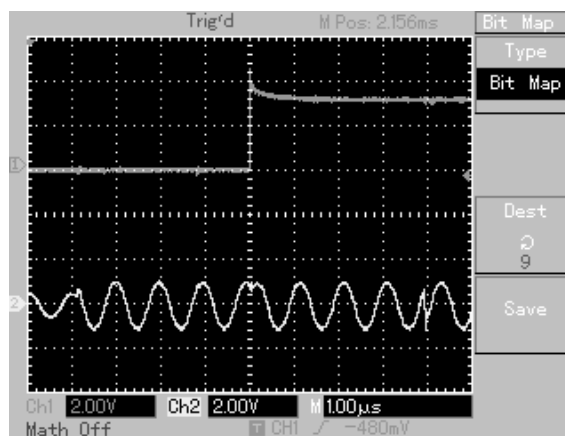
**FIG. 5.4 (c)**

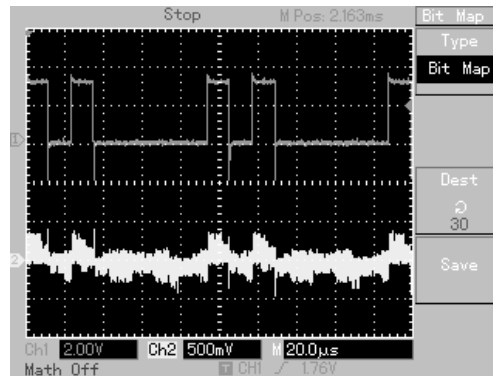
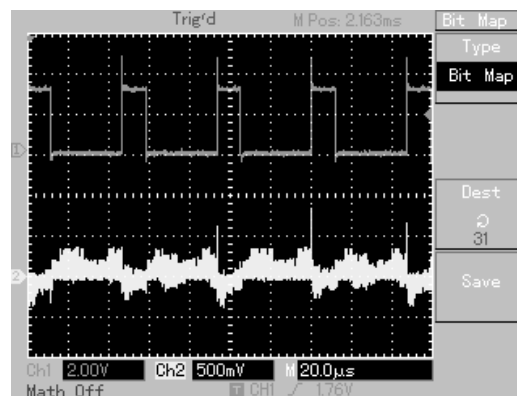
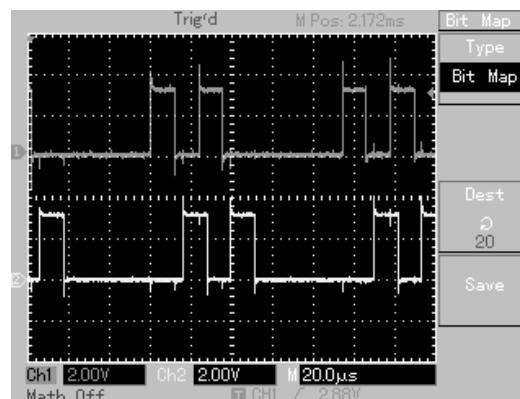
CH 1: EVEN CLK & CH 2: EVEN (TRIBIT ENCODER)**FIG. 5.4 (d)****CH 1: ODD CLK & CH 2: ODD (TRIBIT ENCODER)****FIG. 5.4 (e)**

CH 1: C CLK & CH 2: C (TRIBIT ENCODER)**FIG. 5.4 (f)****CH 1: EVEN (TRIBIT ENCODER) & CH 2: OUT1****Fig.5.4 (g)****CH 1: ODD (TRIBIT ENCODER) & CH 2: OUT2****Fig.5.4 (h)**

CH 1: SIN 1 (0°) & CH 2: SIN 2 (90°)**Fig.5.4 (i)****CH 1: SIN 3 (180°) & CH 2: SIN 4 (270°)****Fig.5.4 (j)****NOTE:**

For **MOD OUT**, select a symbol using **EVEN** and **ODD** and then observe the **MOD OUT** for waveform below a symbol is selected using **EVEN** and **ODD** and then **MOD OUT** is observed with respect to **EVEN**.

CH 1: OUT1 & CH 2: MOD OUT**Fig.5.4(k)**

CH 1:OUT1 & CH 2: MID EVEN**Fig.5.4 (l)****CH 1: OUT2& CH 2: MID ODD****Fig.5.4 (m)****CH 1: OUT1 & CH 2: EVEN (Demodulated)****Fig.5.4 (n)**

CH 1: OUT2 & CH 2: ODD (Demodulated)

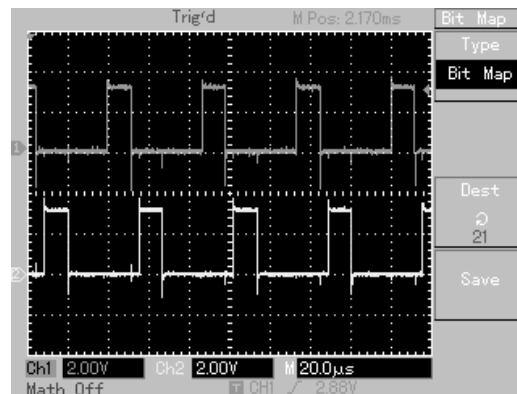


Fig.5.4 (o)

CH 1: C (TRIBIT ENCODER) & CH 2: C(Demodulated)

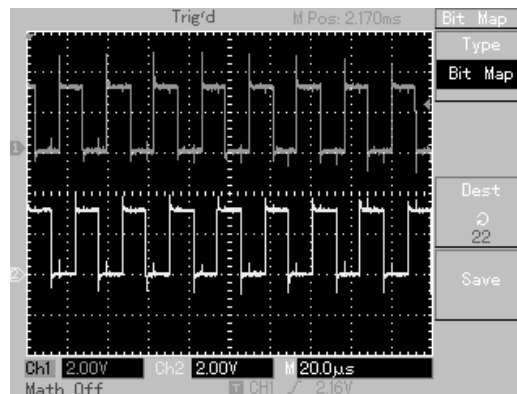


Fig.5.4 (p)

CH 1: EVEN (TRIBIT ENCODER) & CH 2: OUT3

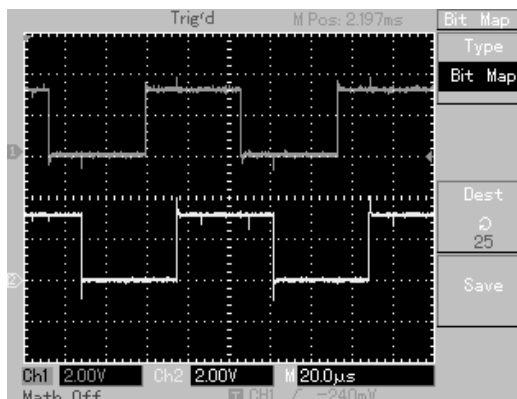
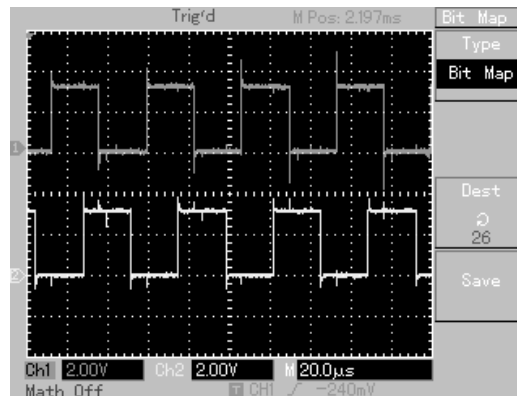
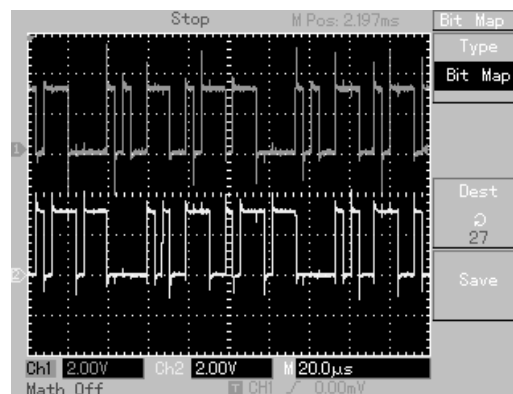


Fig.5.4 (q)

CH 1: ODD (TRIBIT ENCODER) & CH 2: OUT4**Fig.5.4 (r)****CH 1: SERIAL DATA & CH 2: DATA OUT****Fig.5.4 (s)****CONCLUSION:**

The differential coding of EVEN & ODD bit data to be transmitted makes the “bit pair” to be transformed into phase variation of the carrier. In this way the receiver recognizes the pair each time it detects a phase shift of the modulated carrier, independently from its absolute phase. The ambiguity of the QAM modulation, which can take to the demodulated data inversion, is solved in this way.